Set 1: What people mean by "AI"

Outcome AI.BASICS.01: Describe AI in plain language as computer systems that perform tasks such as recognizing, predicting, recommending or generating, and recognize "AI" as a broad umbrella term

Module: Foundation 1: What AI is and where it appears

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
ARTIFICIAL INTELLIGENCE (AI) is a broad name for computer systems that do tasks we usually connect with human thinking, such as recognizing speech, sorting photos, predicting what comes next, recommending a song, or writing a paragraph. It is an umbrella term, not one machine: a spam filter, a map app's travel-time estimate and a chatbot can all be called AI, yet they work differently and do different jobs. Calling something "AI" does not tell you how well it works or what it can do. To understand a tool, ask what task it performs, what it takes in, and what it gives back.

Which description fits this course's meaning of AI?

A. Any computer program at all, including a basic pocket calculator
B. A single website that answers questions in a chat window
C. One specific robot that thinks in exactly the same way as a person
D. A broad name for computer systems that do tasks linked with human thinking

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q1
Artificial Intelligence, Set 1: What people mean by "AI" · MAIN · Q1/5 · Outcome AI.BASICS.01 (local)

Learn
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A phone groups photos by the faces in them, a music app suggests songs, and a chatbot writes a poem. Which statement is accurate?

A. All three must run on the very same AI system inside
B. Only the chatbot counts as AI, because it writes words
C. All three can be called AI, but they do different jobs
D. None of them counts as AI, since each does only one task

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q2
Artificial Intelligence, Set 1: What people mean by "AI" · MAIN · Q2/5 · Outcome AI.BASICS.01 (local)

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A product box says "Now with AI!" What does that label tell you by itself?

A. That the product now thinks the way a person thinks
B. Only that some feature uses AI, not how well it works
C. That it is more accurate than every older version
D. That every feature in the product learned from data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q3
Artificial Intelligence, Set 1: What people mean by "AI" · MAIN · Q3/5 · Outcome AI.BASICS.01 (local)

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Which three questions does the passage suggest asking about any AI tool?

A. What task it does, what it takes in, and what it gives back
B. How popular it is, how new it is, and who recommends it
C. Who invented it, what year it came out, and what it costs
D. Whether it has a robot body, a voice and a face

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q4
Artificial Intelligence, Set 1: What people mean by "AI" · MAIN · Q4/5 · Outcome AI.BASICS.01 (local)

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A spam filter moves suspicious emails into a junk folder. Which AI task is it doing?

A. Recommending new contacts to add
B. Writing new emails for the user
C. Translating emails into other languages
D. Sorting each email into a category

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q5
Artificial Intelligence, Set 1: What people mean by "AI" · MAIN · Q5/5 · Outcome AI.BASICS.01 (local)

Learn
Think of the word VEHICLE. A bicycle, a bus and a cargo ship are all vehicles, but you would not expect a bicycle to cross an ocean. "Vehicle" names a broad group, not one machine, and knowing that something is a vehicle tells you little about its speed or range. "AI" works the same way. It names a broad group of computer systems that do tasks such as recognizing, predicting, recommending or generating. Knowing that something is "AI" tells you little about what it can do. You learn more by asking what job it does.

Why is "vehicle" a good comparison for "AI" in the passage?

A. Both words describe ways of moving people around
B. Both words describe one machine with many parts
C. Both words name a broad group of different things
D. Both words describe machines that have engines

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q1
Artificial Intelligence, Set 1: What people mean by "AI" · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.BASICS.01 (local)

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Knowing that a tool is "AI" is like knowing that something is "a vehicle." What does that alone tell you?

A. Very little about what it can actually do
B. That it can do any job that a person can
C. Exactly how accurate and how fast it is
D. That it is safe to use for any purpose

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q2
Artificial Intelligence, Set 1: What people mean by "AI" · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.BASICS.01 (local)

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Which question tells you more about an AI tool?

A. Is it labeled "AI"?
B. What job does it do?
C. Is it a new release?
D. Is its logo modern?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q3
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A bus and a bicycle are both vehicles. A chatbot and a photo-tagging tool are both what?

A. AI systems that do different jobs
B. chatbots that only write text
C. tools that cannot be called AI
D. the same AI system with two names

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q4
Artificial Intelligence, Set 1: What people mean by "AI" · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.BASICS.01 (local)

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Which task fits the passage's list of AI tasks (recognizing, predicting, recommending, generating)?

A. Printing the page you selected
B. Turning the screen brightness up
C. Suggesting a movie you might enjoy
D. Saving a file when you click Save

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q5
Artificial Intelligence, Set 1: What people mean by "AI" · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.BASICS.01 (local)

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Lena's school announces an "AI tutor." What is the most useful first question to ask?

A. Is it the same AI as every chatbot?
B. What exactly will it help students do?
C. Which company made its logo and colors?
D. Is it smarter than all the teachers?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q1
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.BASICS.01 (local)

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A store's website says its prices are "set by AI." Which statement is accurate?

A. A robot employee chooses each price by hand
B. A chatbot writes out every price for the store
C. The prices must be the lowest prices available anywhere
D. Some system helps set prices; the label alone does not say how

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q2
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.BASICS.01 (local)

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Marcus says, "My map app isn't AI because it doesn't talk to me." What is wrong with his reasoning?

A. AI also includes systems that predict, like travel times

B. Map apps are never allowed to use any kind of AI

C. AI means a robot, so an app can never qualify

D. Nothing; only talking systems can be called AI at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q3
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.BASICS.01 (local)

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A job ad asks for "experience with AI." What should an applicant clarify?

A. Whether the job needs building a human mind
B. Whether the employer owns a humanlike robot
C. Whether AI is one program that everyone uses
D. Which AI tools or tasks the job actually involves

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q4
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.BASICS.01 (local)

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Two friends argue about whether a smart thermostat that learns your schedule is AI. Which response fits the passage best?

A. It counts as AI only if it is as smart as a person

B. It cannot be AI because it has no screen

C. It can be called AI; what matters is the task it performs

D. It counts as AI only if it can hold a conversation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q5
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.BASICS.01 (local)

Learn
Because "AI" is a broad label, it is used loosely. Marketers may call a simple fixed rule "AI" to sound modern, and some tools that do use learned patterns never mention AI at all. Two products both described as "AI-powered" can differ greatly in quality, cost and risk. When a claim matters, for example when a school, employer or bank says AI helps make a decision, look past the label: ask what the system does, what information it uses, how it was tested, and who checks its results. A label is a starting point for questions, not an answer.

A budgeting app calls itself "AI-powered," but it only applies one fixed rule: save 10 percent of each deposit. What is the best conclusion?

A. It is more reliable because it says "AI-powered"

B. The label may be marketing; a fixed rule is doing the work

C. It must be learning from millions of other users' data

D. It cannot work, since it is not "real" AI

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q1
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.BASICS.01 (local)

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Two résumé-screening tools are both described as "AI-powered." What should a hiring manager compare?

A. What each checks, how it was tested, and who reviews results
B. Nothing, since the same label means the same quality
C. Which one has the newer logo and website
D. Which one uses the word "AI" more often

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q2
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.BASICS.01 (local)

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A bank says AI helps decide loan applications. Which question looks past the label?

A. Does the AI have a friendly name and avatar?
B. Is this year's AI newer than last year's AI?
C. What information is used, and can a person review it?
D. Is it the same AI that powers popular chatbots?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q3
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.BASICS.01 (local)

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A photo app's background-blur feature never mentions AI, but it learned from many example photos. Which statement is accurate?

A. It cannot use AI, because AI is never mentioned
B. It must secretly be a chatbot underneath
C. Only features that are advertised can learn from data
D. It may use AI even though it does not say so

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q4
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A news story says a city "used AI" to plan bus routes. What would you need to know before judging the plan?

A. Whether the city also uses AI for its parking meters
B. What the system did, what data it used, and how it was checked
C. Whether the headline used the word "AI" in large type
D. Whether AI is a popular topic in the news this year

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q5
Artificial Intelligence, Set 1: What people mean by "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.BASICS.01 (local)

Set 2: Written rules versus learned patterns

Outcome AI.BASICS.02: Tell software that follows rules a person wrote from a system whose behavior comes from patterns learned from examples

Module: Foundation 1: What AI is and where it appears

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Learn
Some software follows RULES that a person wrote: "If the total is over 50 dollars, apply free shipping." Every step was decided in advance, so you can read the rule to see why the program acted. Other systems LEARN PATTERNS from examples. A spam filter shown thousands of emails marked "spam" or "not spam" adjusts itself until it sorts new emails well. No one wrote a rule for every case; its behavior comes from the examples it learned from. Rule-based software is predictable but cannot handle cases its rules miss. Learned systems handle messy, varied input but can make surprising mistakes.

A checkout page gives free shipping whenever an order is over 50 dollars. How was this behavior most likely produced?

A. It learned from many past orders
B. It copied what a chatbot suggested
C. A person wrote a rule for it
D. It guessed the amount at random

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q1
Artificial Intelligence, Set 2: Written rules versus learned patterns · MAIN · Q1/5 · Outcome AI.BASICS.02 (local)

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A spam filter was shown thousands of emails labeled "spam" or "not spam" and now sorts new emails. Where does its behavior mainly come from?

A. A list of every spam email ever sent

B. The user approving each email by hand first

C. A rule written for every possible email

D. Patterns learned from the labeled examples

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q2
Artificial Intelligence, Set 2: Written rules versus learned patterns · MAIN · Q2/5 · Outcome AI.BASICS.02 (local)

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Which is an advantage of rule-based software?

A. You can read the rule to see why it acted
B. It never needs anyone to update it over time
C. It handles any situation, even ones nobody planned
D. It improves on its own as it sees more data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q3
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Which is a common weakness of a system that learned from examples?

A. It can never handle messy or varied input
B. It can make surprising mistakes on unusual inputs
C. It must be rewritten by hand for each new case
D. It can follow only one fixed rule at a time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q4
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A thermostat turns the heat on at exactly 6:00 a.m. every weekday. Which describes it?

A. A chatbot predicting the weather each day
B. A model trained on thousands of houses
C. A written rule that runs on a schedule
D. A learned pattern from the family's habits

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-M-Q5
Artificial Intelligence, Set 2: Written rules versus learned patterns · MAIN · Q5/5 · Outcome AI.BASICS.02 (local)

Learn
Imagine two cooks. The first follows a RECIPE exactly: "Add 2 teaspoons of salt." If the recipe is right, the soup is right, and if anyone asks why, the cook points to the recipe. The second cook LEARNED BY TASTING hundreds of soups and adjusting until they tasted good. She can handle new ingredients the recipe never mentions, but she cannot always explain exactly why she added what she did, and sometimes she gets a new dish wrong. Rule-based software is like the recipe cook. Systems that learn patterns from examples are like the tasting cook.

Which cook is like rule-based software?

A. The one who follows the recipe exactly
B. The one who learned by tasting many soups
C. The one who refuses to cook new dishes
D. The one who changes the recipe at random

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q1
Artificial Intelligence, Set 2: Written rules versus learned patterns · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.BASICS.02 (local)

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Why can the recipe cook explain her soup easily?

A. She tasted hundreds of soups first
B. She can point to the written steps
C. She guesses differently each time
D. She asks the diners to decide

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q2
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What can the tasting cook do that the recipe cook cannot?

A. Follow a written recipe exactly, step by step
B. Get every new dish right every time
C. Always explain each choice exactly
D. Handle ingredients the recipe never mentions

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What is a weakness of the tasting cook?

A. She cannot handle any new ingredient
B. She needs the recipe for every step
C. She sometimes gets a new dish wrong
D. She never adjusts what she does

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Artificial Intelligence, Set 2: Written rules versus learned patterns · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.BASICS.02 (local)

Learn
Imagine two cooks. The first follows a RECIPE exactly: "Add 2 teaspoons of salt." If the recipe is right, the soup is right, and if anyone asks why, the cook points to the recipe. The second cook LEARNED BY TASTING hundreds of soups and adjusting until they tasted good. She can handle new ingredients the recipe never mentions, but she cannot always explain exactly why she added what she did, and sometimes she gets a new dish wrong. Rule-based software is like the recipe cook. Systems that learn patterns from examples are like the tasting cook.

A system learned to recognize birds from thousands of labeled photos. Which cook is it like?

A. Neither cook
B. Both cooks equally
C. The recipe cook
D. The tasting cook

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-B1-Q5
Artificial Intelligence, Set 2: Written rules versus learned patterns · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.BASICS.02 (local)

Learn
Some software follows RULES that a person wrote: "If the total is over 50 dollars, apply free shipping." Every step was decided in advance, so you can read the rule to see why the program acted. Other systems LEARN PATTERNS from examples. A spam filter shown thousands of emails marked "spam" or "not spam" adjusts itself until it sorts new emails well. No one wrote a rule for every case; its behavior comes from the examples it learned from. Rule-based software is predictable but cannot handle cases its rules miss. Learned systems handle messy, varied input but can make surprising mistakes.

A library wants to charge 25 cents per day for each overdue book. Which approach fits best?

A. A written rule, because the policy is exact and simple
B. A chatbot that guesses a fair fee for each reader
C. A learned model trained on years of borrowing data
D. A model that learns each librarian's fee habits

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q1
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.BASICS.02 (local)

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An app must recognize handwritten numbers written by thousands of different people. Which approach fits best?

A. A fixed rule checking that every 7 has a line
B. Learning patterns from many labeled examples
C. Asking users to type each number instead
D. One written rule for each person's handwriting

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q2
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.BASICS.02 (local)

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A school rule says any student absent 3 days in a row gets a call home. Which approach should run it?

A. A model that learns from past principals' choices
B. A chatbot that decides case by case each morning
C. A written rule that checks for 3 days in a row
D. A model trained to guess which families want calls

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q3
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.BASICS.02 (local)

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A photo service wants to suggest "beach" or "forest" tags for vacation pictures. Which approach fits best?

A. A model that learned from many labeled photos
B. A rule that marks every photo taken in July "beach"
C. A rule that checks whether a file name says "beach"
D. Asking a person to tag every photo by hand forever

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q4
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.BASICS.02 (local)

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Which situation most needs the predictability of written rules?

A. Sorting thousands of customer comments by topic
B. Recognizing which animal appears in a photo
C. Suggesting songs a listener might enjoy next
D. Calculating sales tax that the law sets as an exact rate

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F1-Q5
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.BASICS.02 (local)

Learn
Many real products mix both approaches. A learned model might flag a bank transaction as unusual, and then a written rule decides what happens next, such as "hold any flagged payment over 1,000 dollars." When something goes wrong, the fix depends on which part failed. A wrong written rule can be found and rewritten. A learned pattern usually changes only by retraining with better or more examples, and its mistakes can be harder to explain. Rules give predictability where it matters most; learning helps with variety that rules cannot list in advance.

A bank's model flags a normal grocery purchase as unusual, and a written rule then holds the payment. Which part made the first mistake?

A. The grocery store's card reader
B. The learned model that flagged the purchase
C. The written rule that held the payment
D. The customer who bought the groceries

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q1
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.BASICS.02 (local)

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A written rule says "hold flagged payments over 100 dollars," but the bank meant 1,000. How can this be fixed?

A. Retrain the model on more transactions
B. Wait for the rule to learn the right amount
C. Remove the learned model from the system
D. Rewrite the rule with the correct amount

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q2
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.BASICS.02 (local)

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A learned model keeps misreading a new style of receipt. What is the usual way to improve it?

A. Make the receipts look like old ones
B. Tell it the correct answer once in a chat
C. Retrain it with examples of the new receipts
D. Edit one line of its rules by hand

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q3
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.BASICS.02 (local)

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Why might a designer use a written rule for the final payment decision even when a model flags cases?

A. Rules handle every possible case automatically
B. Rules give predictable results where it matters most
C. Rules never need to be checked or updated
D. Rules can recognize unusual payments by themselves

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q4
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.BASICS.02 (local)

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A customer asks why their payment was held. Which part is easier to explain?

A. The written rule about which flagged payments are held
B. Both parts are always equally easy to explain
C. The exact reason the model flagged the payment
D. Neither part can ever be explained at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S2-F2-Q5
Artificial Intelligence, Set 2: Written rules versus learned patterns · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.BASICS.02 (local)

Set 3: Machine learning and deep learning

Outcome AI.BASICS.03: Explain machine learning and deep learning as nested ideas, with deep learning using many-layered neural networks

Module: Foundation 1: What AI is and where it appears

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
MACHINE LEARNING is a way of building AI in which a system improves at a task by learning patterns from data instead of following only hand-written rules. DEEP LEARNING is one kind of machine learning. It uses NEURAL NETWORKS: many connected layers of simple math units. Each layer passes numbers to the next, and early layers pick up simple features while later layers combine them into more complex ones. "Deep" means many layers, not deep understanding. So the ideas nest like boxes: deep learning sits inside machine learning, which sits inside AI. Not all AI uses machine learning, and not all machine learning is deep learning.

How are AI, machine learning and deep learning related?

A. They are three different words for exactly the same thing
B. AI is part of deep learning, which is part of machine learning
C. Deep learning is part of machine learning, which is part of AI
D. Machine learning and deep learning are unrelated to AI

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q1
Artificial Intelligence, Set 3: Machine learning and deep learning · MAIN · Q1/5 · Outcome AI.BASICS.03 (local)

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What does "deep" mean in "deep learning"?

A. The training data is kept hidden
B. The network has many layers
C. The system understands ideas deeply
D. The model is very old and stable

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q2
Artificial Intelligence, Set 3: Machine learning and deep learning · MAIN · Q2/5 · Outcome AI.BASICS.03 (local)

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What is the key idea of machine learning?

A. A person writes a separate rule for every possible case
B. A program stores every answer and repeats it
C. A computer copies human brain cells into a chip
D. A system improves at a task by learning patterns from data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q3
Artificial Intelligence, Set 3: Machine learning and deep learning · MAIN · Q3/5 · Outcome AI.BASICS.03 (local)

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In a neural network, what do the layers do?

A. Pass numbers along, turning simple features into complex ones
B. Connect the computer to the internet to look up answers
C. Store a separate copy of every training example
D. Hold the written rules that a programmer typed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q4
Artificial Intelligence, Set 3: Machine learning and deep learning · MAIN · Q4/5 · Outcome AI.BASICS.03 (local)

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Which statement is accurate?

A. All AI is deep learning by definition
B. Deep learning is a separate field from AI
C. Some AI uses no machine learning at all
D. All machine learning uses deep neural networks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-M-Q5
Artificial Intelligence, Set 3: Machine learning and deep learning · MAIN · Q5/5 · Outcome AI.BASICS.03 (local)

Learn
Picture three boxes that fit inside each other. The biggest box is labeled ARTIFICIAL INTELLIGENCE: any computer system doing thinking-like tasks. Inside it sits a smaller box, MACHINE LEARNING: AI that learns patterns from data. Inside that sits the smallest box, DEEP LEARNING: machine learning that uses neural networks with many layers. Everything in a small box is also in the boxes around it, but the big box holds other things too. A chess program built only from hand-written rules sits in the AI box but outside the machine learning box.

Which is the biggest box?

A. Artificial intelligence
B. Machine learning box
C. Deep learning
D. Neural network

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q1
Artificial Intelligence, Set 3: Machine learning and deep learning · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.BASICS.03 (local)

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Where does deep learning sit?

A. Beside AI, not in it
B. Around machine learning
C. Outside the AI box
D. Inside machine learning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q2
Artificial Intelligence, Set 3: Machine learning and deep learning · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.BASICS.03 (local)

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A deep learning system is in which boxes?

A. None of the boxes
B. All three boxes
C. Only the AI box
D. Only the deep learning box

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q3
Artificial Intelligence, Set 3: Machine learning and deep learning · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.BASICS.03 (local)

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A chess program built only from hand-written rules is in which box?

A. The machine learning box
B. The deep learning box
C. No box at all
D. The AI box only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q4
Artificial Intelligence, Set 3: Machine learning and deep learning · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.BASICS.03 (local)

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What puts something in the machine learning box?

A. It learns patterns from data
B. It has many hand-written rules
C. It runs on a fast computer
D. It has a robot body

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-B1-Q5
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A tool sorts emails using only rules a person wrote, such as "if the subject contains FREE, mark as junk." Which label fits best?

A. Deep learning with many network layers

B. Not AI, because it cannot write emails

C. AI built from rules, not machine learning

D. Machine learning trained on labeled emails

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q1
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An app recognizes plant species from photos using a neural network with dozens of layers. Which label is most specific?

A. Not AI, since it only looks at pictures
B. Deep learning
C. Machine learning only, not deep learning
D. Rule-based AI without learning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q2
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.BASICS.03 (local)

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A shop predicts next week's sales from past weekly sales using a simple formula fitted to the data, with no neural network. Which label fits?

A. Machine learning that is not deep learning
B. Deep learning, since it learns from data
C. Rule-based AI written by hand
D. Not AI, since it predicts numbers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q3
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.BASICS.03 (local)

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Someone says, "It uses deep learning, so it truly understands." What is the best response?

A. That is right; deep systems understand meaning
B. Deep learning cannot be used for language tasks
C. That is right; neural networks are real brains
D. "Deep" means many layers, not understanding

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q4
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.BASICS.03 (local)

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Which example sits in the machine learning box but not the deep learning box?

A. A list of rules typed in by a programmer by hand
B. A calculator adding two numbers
C. A model fitted to past data with no neural network
D. A neural network with forty layers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F1-Q5
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.BASICS.03 (local)

Learn
People often use these terms loosely, so read claims carefully. "Neural network" borrows a word from biology, but these networks are math, not brains: the comparison helped early researchers and can mislead beginners. "Deep" counts layers; it does not mean wise or thoughtful. Deep learning often needs much more data and computing power than simpler machine learning, and simpler methods can work as well on small, tidy problems such as predicting sales from a few numbers. Choosing an approach depends on the task, the data available and the need to explain results.

An ad says its "neural network works just like a human brain." What is the problem with this claim?

A. Neural networks cannot be used in any products
B. Neural networks are math inspired by a biology word, not brains
C. Neural networks are physically built from brain cells
D. Nothing; that comparison is scientifically exact

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q1
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.BASICS.03 (local)

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A small business has 200 rows of tidy sales numbers and wants a simple forecast it can explain. What does the passage suggest?

A. Deep learning is always required for any forecast
B. Forecasting is impossible without a neural network
C. A simpler method may work as well and be easier to explain
D. More layers always produce more accurate forecasts

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q2
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.BASICS.03 (local)

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Which is a common cost of deep learning compared with simpler methods?

A. It often needs more data and computing power
B. It must be written by hand line by line
C. It can only handle a few numbers at a time
D. It cannot learn from examples at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q3
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.BASICS.03 (local)

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A headline says, "Deep AI now thinks deeply about your questions." What should a careful reader conclude?

A. The system has been proven to reflect like a person
B. "Deep" means the system was trained for a long time
C. All AI is now deep learning, so it thinks
D. The headline mixes up "deep" (layers) with deep thought

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q4
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A team must explain each decision to customers and has little data. Which consideration matters most when choosing an approach?

A. Whether the method is described as "deep"
B. Whether the method can be explained with this data
C. Whether competitors call their tools neural
D. Whether the method has the most layers possible

REFERENCE LABEL: NOT PART OF THE QUESTION ID S3-F2-Q5
Artificial Intelligence, Set 3: Machine learning and deep learning · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.BASICS.03 (local)

Set 4: Four everyday AI jobs

Outcome AI.BASICS.04: Classify an AI use as recommending, recognizing, forecasting or generating

Module: Foundation 1: What AI is and where it appears

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Most everyday AI does one of four kinds of jobs. RECOMMENDING suggests items you might want, such as videos, products or friends. RECOGNIZING (also called classifying) puts input into categories: a face in a photo, a spoken word, a fraudulent payment. FORECASTING predicts a number or event, such as tomorrow's temperature, travel time or next month's sales. GENERATING creates new content: text, images, audio or code. One app can do several jobs, for example recognizing your voice and then generating a reply. Naming the job helps you decide what could go wrong and how to check the result.

A video site shows "Up next" suggestions based on what you watched. Which job is this?

A. Recognizing
B. Forecasting
C. Generating
D. Recommending

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q1
Artificial Intelligence, Set 4: Four everyday AI jobs · MAIN · Q1/5 · Outcome AI.BASICS.04 (local)

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A phone unlocks when it identifies its owner's face. Which job is this?

A. Recommending
B. Recognizing
C. Generating
D. Forecasting

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q2
Artificial Intelligence, Set 4: Four everyday AI jobs · MAIN · Q2/5 · Outcome AI.BASICS.04 (local)

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A weather app predicts a 70 percent chance of rain tomorrow. Which job is this?

A. Recommending
B. Recognizing
C. Forecasting
D. Generating

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q3
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A tool writes a birthday poem from a short description. Which job is this?

A. Generating
B. Recognizing
C. Recommending
D. Forecasting

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q4
Artificial Intelligence, Set 4: Four everyday AI jobs · MAIN · Q4/5 · Outcome AI.BASICS.04 (local)

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A voice assistant hears "What's the weather?" and then writes a spoken reply. Which jobs does it do?

A. Only forecasting what the user will ask next

B. Only recommending a weather app to install

C. Recognizing speech, then generating a reply

D. Only generating; no recognition is needed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-M-Q5
Artificial Intelligence, Set 4: Four everyday AI jobs · MAIN · Q5/5 · Outcome AI.BASICS.04 (local)

Learn
Think of four helpers in a shop. The SALES CLERK says, "People who bought this also liked that": recommending. The CASHIER scans an item and knows it is a banana, not a lemon: recognizing. The MANAGER looks at past weeks and says, "We will sell about 300 loaves on Saturday": forecasting. The SIGN PAINTER creates a brand-new poster for a sale: generating. AI systems do these same four kinds of jobs, just with software instead of people.

The cashier knows a scanned item is a banana. Which job is that?

A. Recognizing
B. Generating
C. Recommending
D. Forecasting

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q1
Artificial Intelligence, Set 4: Four everyday AI jobs · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.BASICS.04 (local)

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The manager predicts Saturday bread sales. Which job is that?

A. Generating
B. Forecasting
C. Recommending
D. Recognizing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q2
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Which helper is generating?

A. The manager
B. The cashier
C. The sales clerk
D. The sign painter

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"People who bought this also liked that" matches which job?

A. Forecasting
B. Generating
C. Recommending
D. Recognizing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q4
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An app predicts how many minutes your bus will take. Which shop helper does the same kind of job?

A. The manager
B. The cashier
C. The sales clerk
D. The sign painter

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-B1-Q5
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A bank text says, "We blocked a payment that looked like fraud." Which job did the AI most likely do?

A. Forecasting next year's interest rates
B. Recognizing the payment as likely fraud
C. Recommending a product to the customer
D. Generating a new payment for the customer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q1
Artificial Intelligence, Set 4: Four everyday AI jobs · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.BASICS.04 (local)

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A delivery app says your order will arrive in 25 minutes. Which job is this?

A. Recognizing
B. Generating
C. Recommending
D. Forecasting

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q2
Artificial Intelligence, Set 4: Four everyday AI jobs · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.BASICS.04 (local)

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A design app makes three new logo drafts from a description. Which job is this?

A. Recognizing
B. Forecasting
C. Generating
D. Recommending

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A shopping site shows "Customers also bought" items. What could go wrong that fits this job?

A. It keeps suggesting more of the same, missing your real need
B. It writes a false product description
C. It mislabels your photo as a cat
D. It miscounts the exact delivery time of your order

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q4
Artificial Intelligence, Set 4: Four everyday AI jobs · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.BASICS.04 (local)

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A camera app labels a photo of a husky as "wolf." Which job made the mistake?

A. Recommending
B. Recognizing
C. Forecasting
D. Generating

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F1-Q5
Artificial Intelligence, Set 4: Four everyday AI jobs · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.BASICS.04 (local)

Learn
Each job fails in its own way, so naming the job tells you what to check. A RECOMMENDER can keep showing more of the same and miss what you actually need. A RECOGNIZER can put things in the wrong category, especially with unusual examples. A FORECAST can miss sudden changes that never appeared in past data. A GENERATOR can produce fluent content that is false or copied. When one app chains jobs together, an early mistake can spread: if your speech is misrecognized, the generated reply answers the wrong question.

A voice app answers "Book a table for four" by booking a table for two. Speech was turned into text, then a booking was made. What should be checked first?

A. Whether the booking page used AI at all
B. Whether the restaurant forecast was wrong
C. Whether the recommendations were too narrow
D. Whether the speech was recognized correctly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q1
Artificial Intelligence, Set 4: Four everyday AI jobs · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.BASICS.04 (local)

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A sales forecast built from three normal years badly misses a month when a new factory opened nearby. Why?

A. Forecasts can miss sudden changes not in past data

B. Recommenders made the forecast too narrow

C. Forecasts always ignore all past data

D. The forecast must have generated false text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q2
Artificial Intelligence, Set 4: Four everyday AI jobs · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.BASICS.04 (local)

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A tool generated a confident product description listing a feature the product does not have. What kind of failure is this?

A. A forecaster missing a sudden change
B. A generator producing fluent but false content
C. A recommender narrowing suggestions
D. A recognizer putting the product in a category

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q3
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A music app plays only songs very similar to ones you already liked. What is the best fix to try?

A. Retrain its speech recognition to understand your voice

B. Ask it to generate brand-new songs instead

C. Check its weather forecast against a website

D. Look for settings or searches that widen the suggestions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q4
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A photo tool identifies plants well in a garden but poorly for rare desert plants. What does this suggest?

A. Forecasts miss sudden changes in sales

B. Generators invent fluent but false text

C. Recognizers struggle with unusual examples

D. Recommenders repeat the same suggestions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S4-F2-Q5
Artificial Intelligence, Set 4: Four everyday AI jobs · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.BASICS.04 (local)

Set 5: Generative versus predictive uses

Outcome AI.BASICS.05: Decide whether a task needs new content generated or a label or number predicted, and name a typical failure of each

Module: Foundation 1: What AI is and where it appears

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Two broad uses of AI are easy to confuse. PREDICTIVE (or classifying) uses produce a LABEL or a NUMBER about the input: "spam or not spam," "92 percent likely to be a cat," "about 30 minutes." GENERATIVE uses produce NEW CONTENT: a paragraph, a picture, a song, a block of code. They fail differently. A prediction can be the wrong label or a number that is off. Generated content can look polished while containing false facts, invented details or material too close to someone else's work. Before using AI, ask: do I need a label or a number, or do I need new content?

A school wants each incoming email marked "absence," "question" or "other." Which kind of AI use is this?

A. Predictive: it produces a label
B. Generative: it writes new emails
C. Generative: it creates new categories each day
D. Neither; labeling emails cannot use AI

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q1
Artificial Intelligence, Set 5: Generative versus predictive uses · MAIN · Q1/5 · Outcome AI.BASICS.05 (local)

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A student asks a tool for a short story about a lost robot. Which kind of use is this?

A. Predictive: it produces a label
B. Neither; stories cannot be produced by AI
C. Predictive: it produces one number
D. Generative: it produces new content

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q2
Artificial Intelligence, Set 5: Generative versus predictive uses · MAIN · Q2/5 · Outcome AI.BASICS.05 (local)

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Which output is typical of a predictive use?

A. A paragraph of product description
B. A drawing of a mountain at sunset
C. "87 percent likely to be fraud"
D. A new poem about the ocean

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q3
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What is a typical failure of generated content?

A. It returns a predicted number that is a bit too high
B. It looks polished but contains false or invented details
C. It refuses to produce any output at all
D. It gives a single wrong yes or no label

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A clinic wants to estimate how many patients will arrive tomorrow. Which kind of use fits?

A. Generative: a new image
B. Generative: a song about patients
C. Generative: a written story
D. Predictive: a number

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-M-Q5
Artificial Intelligence, Set 5: Generative versus predictive uses · MAIN · Q5/5 · Outcome AI.BASICS.05 (local)

Learn
A useful skill is looking at an OUTPUT and asking what kind it is. A LABEL names a category: "spam," "cat," "approved." A NUMBER measures or predicts something: "42 minutes," "12 percent chance." NEW CONTENT is something made that did not exist before: an essay, a photo, a melody, a joke. Some outputs combine kinds: "Spam (95 percent sure)" is a label plus a number, but it is still not new content. If the output could be checked by asking "which category?" or "how much?", it is a label or a number.

Which output is a label?

A. A photo of a sunset
B. "Approved"
C. A three-paragraph essay
D. A new song melody

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q1
Artificial Intelligence, Set 5: Generative versus predictive uses · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.BASICS.05 (local)

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Which output is a number?

A. "12 percent chance of rain"
B. "Spam"
C. "A new drawing of clouds"
D. "A poem about rain"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-B1-Q2
Artificial Intelligence, Set 5: Generative versus predictive uses · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.BASICS.05 (local)

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Which output is new content?

A. "Cat (label)"
B. "Not spam"
C. A newly written joke
D. "About 42 minutes"

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"Spam (95 percent sure)" is which kind of output?

A. An essay
B. A label plus a number
C. New content only, like an essay
D. A number only

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Which question checks a label?

A. "Does the song rhyme?"
B. "How long is the story?"
C. "Which category is it?"
D. "Is the drawing pretty?"

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A store wants to know which product reviews are positive or negative. Which use fits best?

A. Predictive: label each review as positive or negative
B. Generative: write a poem about the reviews
C. Generative: write new reviews for each product
D. Generative: create images of happy customers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q1
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A teacher wants five new practice questions about fractions. Which use fits best?

A. Predictive: score one answer as right or wrong
B. Predictive: label existing questions by topic
C. Predictive: forecast next week's test scores
D. Generative: create new questions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q2
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.BASICS.05 (local)

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A tool generated five practice questions. What should the teacher check before using them?

A. That a label was predicted for each question
B. That the questions are as long as possible
C. That each question and answer is actually correct
D. Nothing, because generated text is checked already

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q3
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.BASICS.05 (local)

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A city wants an estimate of how many buses are needed at 8 a.m. Which use fits best?

A. Generative: compose a bus jingle
B. Predictive: estimate a number
C. Generative: draw a picture of a bus
D. Generative: write a story about commuters

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q4
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.BASICS.05 (local)

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A volunteer group wants a friendly thank-you letter for donors. Which use fits best?

A. Predictive: mark the letter "approved"
B. Predictive: estimate next year's donations
C. Predictive: label each donor by city
D. Generative: draft a new letter

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F1-Q5
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.BASICS.05 (local)

Learn
Many real tasks mix the two uses. A help desk might first PREDICT which category a question belongs to, then GENERATE a draft reply. A mistake in the first step can make the second step confidently wrong. Some tasks are better kept predictive even when a generator could do them: if you need a yes or no answer, a confidence score and an audit trail, a classifier is easier to test than free text. And when facts matter, generated content needs checking against a source, because polished wording is not evidence.

A help desk tool labels a billing question as "shipping," then drafts a confident reply about delivery times. What went wrong?

A. The wrong label led to a confidently wrong draft
B. The label was right, but the draft was too short
C. Generators can never write help desk replies
D. The draft was right; billing and shipping are the same

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q1
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.BASICS.05 (local)

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A company needs a yes or no decision on each expense claim, with a score and a record it can audit. Which approach is easier to test?

A. A generator that writes a paragraph about each claim
B. A classifier that outputs a label and a score
C. A generator that writes a poem about expenses
D. A generator asked to "decide fairly" in prose

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q2
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.BASICS.05 (local)

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A generated travel guide says a museum is free on Mondays. What should you do?

A. Trust it because the writing is polished and clear
B. Assume all museums are free on Mondays
C. Check the museum's official site before you rely on it
D. Ask the same tool to repeat the claim

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q3
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.BASICS.05 (local)

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A school wants AI to sort essays by topic and then write feedback. Which step should be checked first if the feedback seems off-topic?

A. The sorting step that labeled each essay's topic
B. The length of each feedback paragraph
C. Whether essays should ever get written feedback
D. The paper the essays were printed on

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q4
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.BASICS.05 (local)

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A manager asks a generator, "Is this invoice fraudulent?" and gets a long, confident paragraph. What is the main concern?

A. Generators always refuse fraud questions
B. Long answers are always more accurate
C. The long paragraph proves that the invoice is fraud
D. Confident prose is not evidence the judgment is right

REFERENCE LABEL: NOT PART OF THE QUESTION ID S5-F2-Q5
Artificial Intelligence, Set 5: Generative versus predictive uses · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.BASICS.05 (local)

Set 6: The model versus the product

Outcome AI.BASICS.06: Tell an AI model from the app or service built around it, and avoid assuming that all AI systems or all chatbots behave the same way

Module: Foundation 1: What AI is and where it appears

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A MODEL is the trained part of an AI system: the learned patterns that turn input into output. A PRODUCT (an app, website or feature) wraps a model with other parts: a screen to type in, instructions the company adds, safety filters, tools such as web search, saved settings, and usage limits. The same model can sit inside several products that behave differently, and one product can switch models over time. So "this chatbot did X" does not mean every chatbot does X, and a product's feature, like saving files or browsing the web, may come from the product, not from the model itself.

What is a model in an AI system?

A. The app's screen and buttons you tap
B. The company that sells the subscription
C. The trained part that turns input into output
D. The list of usage limits on an account

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q1
Artificial Intelligence, Set 6: The model versus the product · MAIN · Q1/5 · Outcome AI.BASICS.06 (local)

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Two apps use the same model but behave differently. What is the most likely reason?

A. Each product adds its own instructions, tools and limits
B. The model must be broken in one of the apps
C. Products cannot share a model, so this cannot happen
D. Models change randomly each time you open an app

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q2
Artificial Intelligence, Set 6: The model versus the product · MAIN · Q2/5 · Outcome AI.BASICS.06 (local)

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A chatbot app can search the web. Where does that ability most likely come from?

A. Every model's built-in ability to browse
B. A search tool the product connects to the model
C. The user's own computer searching by itself
D. The model memorizing the entire internet

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q3
Artificial Intelligence, Set 6: The model versus the product · MAIN · Q3/5 · Outcome AI.BASICS.06 (local)

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Your friend says, "Chatbots always remember everything you tell them." What is the best response?

A. Only paid chatbots have models inside
B. That is true of every chatbot ever made
C. No chatbot can store anything at all
D. It depends on the product; features differ

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q4
Artificial Intelligence, Set 6: The model versus the product · MAIN · Q4/5 · Outcome AI.BASICS.06 (local)

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Which is part of the product rather than the model?

A. The patterns learned from example text
B. The learned patterns that produce replies
C. A daily limit on how many messages you can send
D. The numbers set during training

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-M-Q5
Artificial Intelligence, Set 6: The model versus the product · MAIN · Q5/5 · Outcome AI.BASICS.06 (local)

Learn
Think of a car ENGINE and the CAR built around it. The engine makes the power, but the car adds the steering wheel, seats, brakes, radio and speed limiter. The same engine can go into a sports car and a delivery van, and they will drive very differently. An AI MODEL is like the engine: it produces the output. The PRODUCT is like the car: it adds the screen, rules, tools and limits. If a van is slow, the reason might be its limiter, not its engine.

In the comparison, what is the AI model like?

A. The steering wheel
B. The engine
C. The seats
D. The radio

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q1
Artificial Intelligence, Set 6: The model versus the product · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.BASICS.06 (local)

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What is the product like?

A. The car around the engine
B. Only the fuel
C. Only the engine by itself
D. Only the road

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q2
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Why can two cars with the same engine drive differently?

A. Engines change their shape every single day
B. Cars cannot share engines
C. One engine must be broken
D. Their other parts and settings differ

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q3
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A van is slow because of a speed limiter. In AI terms, the slowness comes from what?

A. The user
B. The model
C. The product
D. The training data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-B1-Q4
Artificial Intelligence, Set 6: The model versus the product · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.BASICS.06 (local)

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Which is like a product feature added around the model?

A. The trained numbers
B. A safety filter
C. The model's output
D. The learned patterns

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A school's writing app refuses to write full essays, while another app using the same model will. What explains this?

A. The school's product added its own rules
B. The model in the school app is too weak
C. The other app has a different kind of model
D. Essay writing is impossible for any model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q1
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A coworker says, "The model can read our team calendar." What is more accurate?

A. Calendars are part of every model's patterns
B. Models learn calendars during training
C. Every model can read every calendar automatically
D. The product may connect a calendar tool to the model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q2
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Ana read an article about one chatbot's limits. What should she remember about a different chatbot?

A. It uses no model if it is different
B. It has no limits since it is a different brand
C. Its limits may differ, so check that product
D. Its limits must match the article exactly

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A product updates to a new model but keeps its name. What could change for users?

A. Only the price, never the answers
B. Only the colors of the app
C. Nothing, because the name stayed the same
D. Its answers and abilities could change

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q4
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Which statement about chatbots is accurate?

A. Every chatbot can browse the web and save files
B. Different chatbots can answer one question differently
C. All chatbots give the same answer to any question
D. A chatbot is a model with no product around it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F1-Q5
Artificial Intelligence, Set 6: The model versus the product · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.BASICS.06 (local)

Learn
Because products change, details you learned about one can go out of date: a company may update its model, add or remove tools, or change limits without the name changing much. Different products may also use different kinds of models: a chatbot, a photo tagger and a translation feature are not interchangeable. When a result matters, check what the specific product says about itself today, such as its help pages, settings and release notes, rather than assuming from a friend's experience with a different product or from an old article.

A blog post from two years ago says a product cannot read PDF files. You need to know today. What should you check?

A. The product's current help pages or release notes
B. The same two-year-old blog post again
C. Nothing; products never change their features
D. A friend's experience with another product

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q1
Artificial Intelligence, Set 6: The model versus the product · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.BASICS.06 (local)

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A photo tagger and a chatbot are both called "AI." Can you use the photo tagger to draft a letter?

A. Yes; all AI products can do every task
B. Yes; if both use AI, they must share one model
C. No; letters can never be drafted with AI
D. No; they are different products for different tasks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q2
Artificial Intelligence, Set 6: The model versus the product · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.BASICS.06 (local)

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Your settings page says "Chat history saving: off," but a friend says the product always saves chats. What is the best next step?

A. Delete your account without checking anything
B. Read what the product says the setting does and rely on that
C. Assume your friend is right and ignore the setting entirely
D. Assume no product ever saves anything

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q3
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A product changed its limits last month. Your guide from last year says "unlimited messages." What should you conclude?

A. All guides are always wrong about limits
B. Limits are set by the model, so they never change
C. The guide may be out of date; check the current terms
D. The guide must be right because it was published

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q4
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A company says its product "uses the latest model." Why might that be hard to rely on later?

A. The product may switch models over time
B. Models never change once released
C. "Latest" is a guarantee of accuracy
D. Products cannot contain models

REFERENCE LABEL: NOT PART OF THE QUESTION ID S6-F2-Q5
Artificial Intelligence, Set 6: The model versus the product · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.BASICS.06 (local)

Set 7: Comparing AI with people

Outcome AI.BASICS.07: Use human comparisons carefully: say where a comparison helps, where it misleads, and why abilities can be uneven

Module: Foundation 1: What AI is and where it appears

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People often compare AI with humans: "it reads," "it thinks," "it is like a student." Comparisons can help explain a new idea, but they can also mislead. A system may beat experts at one narrow task, such as spotting a pattern in thousands of images, yet fail at something a child finds easy, like counting objects in an unusual picture. Its abilities can be UNEVEN in ways a human's are not, so success on one task does not predict success on a nearby task. Treat a human comparison as a starting point, then check how the system actually performs on the task you care about.

A system scores better than doctors at spotting one type of spot on X-rays. Which conclusion is justified?

A. It will also be excellent at reading patient notes
B. It can now replace doctors for all medical decisions
C. It must understand medicine as well as a doctor
D. It did well on that narrow task; other tasks need checks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q1
Artificial Intelligence, Set 7: Comparing AI with people · MAIN · Q1/5 · Outcome AI.BASICS.07 (local)

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Why can AI abilities be called "uneven"?

A. It improves evenly like a growing child
B. It may do a hard task well but fail an easy one
C. It always performs the same on every task
D. It only fails at tasks that are very hard

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q2
Artificial Intelligence, Set 7: Comparing AI with people · MAIN · Q2/5 · Outcome AI.BASICS.07 (local)

Learn
People often compare AI with humans: "it reads," "it thinks," "it is like a student." Comparisons can help explain a new idea, but they can also mislead. A system may beat experts at one narrow task, such as spotting a pattern in thousands of images, yet fail at something a child finds easy, like counting objects in an unusual picture. Its abilities can be UNEVEN in ways a human's are not, so success on one task does not predict success on a nearby task. Treat a human comparison as a starting point, then check how the system actually performs on the task you care about.

A teacher says, "The AI is like a student learning to write." What is the best way to use this comparison?

A. As a starting point, then check how it actually performs
B. As a reason never to check its writing
C. As proof that the AI learns exactly like a student
D. As a reason to give it grades like a student

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q3
Artificial Intelligence, Set 7: Comparing AI with people · MAIN · Q3/5 · Outcome AI.BASICS.07 (local)

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A system writes fluent essays but miscounts the apples in a simple picture. What does this show?

A. Counting is always harder than writing for computers
B. The essays must also be full of counting errors
C. Skill at one task does not predict skill at another
D. It is pretending to be bad at counting on purpose

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q4
Artificial Intelligence, Set 7: Comparing AI with people · MAIN · Q4/5 · Outcome AI.BASICS.07 (local)

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Which statement is a careful description rather than a misleading comparison?

A. "It sees the world in just the way we do."
B. "It recognized 95 of 100 test images correctly."
C. "It is as smart as a ten-year-old."
D. "It thinks like a genius."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-M-Q5
Artificial Intelligence, Set 7: Comparing AI with people · MAIN · Q5/5 · Outcome AI.BASICS.07 (local)

Learn
A pocket calculator multiplies huge numbers instantly, faster than any person, yet it cannot tell a cat from a dog. A toddler easily tells a cat from a dog but cannot multiply 27 by 43. Neither is "smarter" overall; they have different abilities. AI systems are similar: a system can be impressive at one thing and weak at another, in a pattern unlike any human's. So ask "How good is it at THIS task?" rather than "How smart is it?"

What can the calculator do better than a person?

A. Tell a cat from a dog at a glance
B. Explain a joke
C. Multiply huge numbers quickly
D. Learn a language

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q1
Artificial Intelligence, Set 7: Comparing AI with people · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.BASICS.07 (local)

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What can the toddler do that the calculator cannot?

A. Add huge numbers
B. Run on batteries
C. Multiply 27 by 43
D. Tell a cat from a dog

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q2
Artificial Intelligence, Set 7: Comparing AI with people · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.BASICS.07 (local)

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Which question does the passage recommend?

A. "How good is it at this task?"
B. "Does it think like us?"
C. "How smart is it overall?"
D. "Is it smarter than a toddler?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q3
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Who is "smarter" overall: the calculator or the toddler?

A. Both are equally smart at everything
B. Neither; they have different abilities
C. The calculator, since it is faster
D. The toddler, since it is human

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q4
Artificial Intelligence, Set 7: Comparing AI with people · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.BASICS.07 (local)

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What does the example suggest about AI?

A. It can be strong at one thing and weak at another
B. It is always stronger than a calculator at math
C. It is equally good at all tasks
D. It is always weaker than a toddler

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-B1-Q5
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A tool translated a menu well. You now want it to translate a medical form. What should you do?

A. Assume medical words are easier to translate than menus
B. Refuse because translation is impossible
C. Trust it fully since the menu was perfect
D. Check its performance on the medical form separately

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q1
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.BASICS.07 (local)

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Jordan says, "This chatbot is basically a person in a box." What is the most accurate reply?

A. Yes, it has the same judgment and common sense as a person
B. No, it can never do anything useful
C. It can sound human, but its abilities differ from a person's
D. Yes, because it uses words like a person

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q2
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.BASICS.07 (local)

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A study tool explains algebra steps clearly but gets one multiplication wrong. What is the best habit?

A. Assume all its explanations are wrong
B. Use the explanation but check each calculation
C. Trust every step because the explanation is clear
D. Stop using it for anything at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q3
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Which report about an AI tool is most useful to a manager?

A. "It is dumb and will never work."
B. "It thinks just like our accountant."
C. "It sorted 180 of 200 invoices; handwritten ones failed."
D. "It is really smart, just like a new hire we trained."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q4
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.BASICS.07 (local)

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An image tool wins an art contest. A friend says it can surely design a safe bridge. What is wrong?

A. Art and bridge design are the same task
B. Image tools cannot win art contests
C. Nothing; winning a contest proves its general skill
D. Success in one task does not show skill in a different one

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F1-Q5
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.BASICS.07 (local)

Learn
Human comparisons shape trust. If a system "sounds like a person," people may assume it has human judgment, feelings or common sense. Some language models are trained to sound helpful and friendly, which can make errors easier to miss. On the other hand, dismissing a system as "just a machine" can hide real abilities and real risks. A better habit is to describe what was observed: "It summarized these ten reports accurately but invented one date," rather than "It is smart" or "It is dumb." Specific observations can be checked.

A chatbot answers warmly, "I totally understand how you feel." What should you keep in mind?

A. Friendly wording does not show feelings or judgment
B. It is lying, so nothing it says is useful
C. Friendly replies are always more accurate than others
D. It must truly share your feelings

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q1
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.BASICS.07 (local)

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Which statement is a checkable observation?

A. "It understands us better than other people do."
B. "It is just a dumb machine."
C. "It is basically genius-level smart."
D. "It summarized ten reports but invented one date."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q2
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.BASICS.07 (local)

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A coworker dismisses a tool as "just a machine" and ignores that it can write convincing fake emails. What is the risk?

A. Calling it a machine makes it safer
B. None, since machines cannot write convincing emails
C. Underestimating real abilities can hide real risks
D. The tool will stop working if ignored

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q3
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Why might a friendly, human-sounding style make errors easier to miss?

A. People tend to trust things that sound like a helper
B. Errors are always marked in red
C. Human-sounding systems cannot make real mistakes
D. Friendly text contains no errors by design

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q4
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.BASICS.07 (local)

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A review says, "The AI is honest and caring." What would make the review more useful?

A. Say it is "better than a human"
B. Describe specific tasks it did well or badly
C. Compare it to a famous person
D. Use stronger words like "loving" and "wise"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S7-F2-Q5
Artificial Intelligence, Set 7: Comparing AI with people · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.BASICS.07 (local)

Set 8: Training versus using a model

Outcome AI.LEARN.01: Tell training (learning patterns from data) from inference (using the finished model on new input)

Module: Foundation 2: How a model learns

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
An AI model has two very different stages. TRAINING is when the model learns: it is shown large amounts of example data and its internal settings are adjusted again and again until its outputs match the examples well. Training happens before you use the model and usually takes a lot of computing power. INFERENCE (or simply "using the model") is when the finished model takes NEW input, such as your question or photo, and produces an output. During ordinary use the model's learned settings do not change. Typing a question into an app is inference, not training.

A company shows a model millions of labeled photos and adjusts it until it sorts them well. Which stage is this?

A. Inference, because the photos are labeled

B. Neither, because sorting photos is not AI

C. Inference

D. Training

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q1
Artificial Intelligence, Set 8: Training versus using a model · MAIN · Q1/5 · Outcome AI.LEARN.01 (local)

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You type a question into a finished chatbot and get a reply. Which stage is this?

A. Training: the model rewrites itself for your question
B. Neither stage, because no model is involved
C. Inference: the finished model handles new input
D. Training: your question becomes part of its settings

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q2
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Which statement about training is accurate?

A. It adjusts the model's settings using many examples
B. It happens each time anyone opens the app on a phone
C. It is the same as typing a question to the model
D. It needs only one example to finish completely

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q3
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During ordinary use (inference), what happens to the model's learned settings?

A. They reset to random values
B. They stay the same
C. They update after every message
D. They copy the user's whole device

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q4
Artificial Intelligence, Set 8: Training versus using a model · MAIN · Q4/5 · Outcome AI.LEARN.01 (local)

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Why does training usually cost much more computing power than a single use?

A. It must print out every training example on paper
B. It only ever handles one short question at a time
C. It repeats adjustments over huge amounts of data
D. It happens on the user's phone after each reply

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-M-Q5
Artificial Intelligence, Set 8: Training versus using a model · MAIN · Q5/5 · Outcome AI.LEARN.01 (local)

Learn
Think about learning to drive. While you STUDY and PRACTICE with a teacher, you make mistakes, get corrected and slowly improve: that is like TRAINING. Once you have your license and drive to a store, you use what you learned on a new road. You are not taking lessons on that trip: that is like INFERENCE. An AI model is similar. It learns during training, which happens first and takes a long time. Later it uses what it learned on new input, quickly, without changing what it learned.

Practicing with a driving teacher is like which stage?

A. Inference
B. Neither stage
C. Both stages equally
D. Training

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q1
Artificial Intelligence, Set 8: Training versus using a model · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LEARN.01 (local)

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Driving to the store with your license is like which stage?

A. Inference
B. Neither stage
C. Training
D. A driving exam

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q2
Artificial Intelligence, Set 8: Training versus using a model · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LEARN.01 (local)

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Which stage usually happens first?

A. Neither happens
B. Training
C. They happen at once
D. Inference

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On the trip to the store, are you taking a lesson?

A. No, you are using what you learned
B. Yes, every trip is a full lesson
C. No, because you forgot how to drive
D. Yes, the store teaches you to drive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q4
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Which stage takes a long time and many corrections?

A. Driving to the store
B. Inference
C. Training
D. Getting the license card

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-B1-Q5
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A hospital's team spends weeks adjusting a model with thousands of past X-rays. Later, a doctor uploads a new X-ray for a second opinion. Which part is inference?

A. Hiring the team that adjusted the model
B. Collecting the thousands of past X-rays
C. The weeks of adjusting the model
D. The doctor uploading the new X-ray

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q1
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Mia asks a translation app to translate a menu. Does her request train the app's model?

A. Yes; every request is a training example instantly
B. No; translating her menu is inference
C. No; translation apps never use any models
D. Yes; the model rewrites its settings for menus

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q2
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A company says its new model "finished training in June." What does that tell you?

A. It cannot answer any questions after June
B. It was trained by answering users' questions in June
C. It keeps training on every question users type
D. Learning from its data ended then; later use is inference

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q3
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Which activity is training?

A. Asking a finished model for a recipe idea
B. Adjusting a model's settings with labeled examples
C. Reading the reply a model already produced
D. Uploading a photo to be tagged by an app

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q4
Artificial Intelligence, Set 8: Training versus using a model · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LEARN.01 (local)

Learn
An AI model has two very different stages. TRAINING is when the model learns: it is shown large amounts of example data and its internal settings are adjusted again and again until its outputs match the examples well. Training happens before you use the model and usually takes a lot of computing power. INFERENCE (or simply "using the model") is when the finished model takes NEW input, such as your question or photo, and produces an output. During ordinary use the model's learned settings do not change. Typing a question into an app is inference, not training.

Why is it useful to know that ordinary use does not change a model's settings?

A. Correcting it in one chat will not fix the model itself
B. It means the model gets smarter with each question you ask
C. It means the model can never be updated by its makers
D. It means the model forgets everything after each reply

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F1-Q5
Artificial Intelligence, Set 8: Training versus using a model · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LEARN.01 (local)

Learn
Products can blur the line between training and use, so it helps to ask exactly what changes. Rating an answer "thumbs up" or "thumbs down" does not instantly retrain the model; a company MAY later use such feedback, along with other data, to train a new version, depending on its policies. Some products keep notes about you between conversations, which feels like learning but is stored information the product feeds back in, not a change to the model's settings. When you want to know whether your data could be used for training, read the provider's data policy and settings.

You click "thumbs down" on a wrong answer. What is the most accurate description?

A. It instantly retrains the model to fix that answer for everyone
B. It deletes the model and replaces it with a better one at once
C. It is feedback the company may use later, depending on its policy
D. It has no possible use for anyone at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q1
Artificial Intelligence, Set 8: Training versus using a model · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LEARN.01 (local)

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A chatbot remembers your name in a new conversation. What is the most likely explanation?

A. The model guessed your name from its training data
B. The product stored your name and fed it back in
C. Every model is born knowing every user's name
D. The model retrained itself on your name overnight

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q2
Artificial Intelligence, Set 8: Training versus using a model · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LEARN.01 (local)

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Sam wants to know whether his uploaded files could be used to train future models. What should he check?

A. Whether the chatbot sounds friendly when he asks
B. Whether the chatbot remembers his name
C. The provider's data policy and his account settings
D. Nothing; uploaded files are never used by anyone

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q3
Artificial Intelligence, Set 8: Training versus using a model · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LEARN.01 (local)

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A user corrects a chatbot's wrong date and it apologizes. Later, a different user asks the same question. What should we expect?

A. The model will now always give the correct date
B. The model will refuse to answer that question again
C. The other user will see the first user's correction
D. The model may make the same mistake for the other user

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q4
Artificial Intelligence, Set 8: Training versus using a model · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LEARN.01 (local)

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Which change would actually alter the model's learned settings?

A. The company training a new version with new data
B. A user rating one reply with a thumbs-up
C. A user typing "remember this" in a chat
D. A user starting a brand-new conversation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S8-F2-Q5
Artificial Intelligence, Set 8: Training versus using a model · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LEARN.01 (local)

Set 9: Data, patterns and parameters

Outcome AI.LEARN.02: Describe parameters as the adjustable numbers a model sets during training, and explain that the model keeps learned patterns, not a copy of every example

Module: Foundation 2: How a model learns

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
During training, a model adjusts a huge set of numbers called PARAMETERS (sometimes called weights). You can picture them as millions or billions of tiny dials. Each time the model's output is wrong on a training example, the dials are nudged a little so that next time the output is closer. After enough nudges, the dial settings capture PATTERNS in the data, for example which pixel shapes usually mean "cat." The model keeps the dial settings, not a neat copy of every example, although it can sometimes reproduce pieces of text or images it saw many times. More parameters can capture more patterns, but size alone does not guarantee better results.

What are parameters in a model?

A. The questions users type into the finished app
B. The list of every training example, kept word for word
C. The rules a programmer wrote out line by line
D. The adjustable numbers set during training

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q1
Artificial Intelligence, Set 9: Data, patterns and parameters · MAIN · Q1/5 · Outcome AI.LEARN.02 (local)

Learn
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What happens to the parameters when the model gets a training example wrong?

A. They are nudged so the next output is closer
B. They are replaced with the example itself
C. They are erased and started over from zero
D. They stay exactly the same as before

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q2
Artificial Intelligence, Set 9: Data, patterns and parameters · MAIN · Q2/5 · Outcome AI.LEARN.02 (local)

Learn
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After training, what does a model mainly keep?

A. A tidy, searchable copy of every example it saw
B. Dial settings that capture patterns in the data
C. Nothing at all; it relearns each time it is used
D. Only the last example it saw during training

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q3
Artificial Intelligence, Set 9: Data, patterns and parameters · MAIN · Q3/5 · Outcome AI.LEARN.02 (local)

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A model has twice as many parameters as another. Which statement is accurate?

A. It cannot be used by ordinary people at all
B. It must have been trained on exactly half the data
C. It can capture more patterns, but bigger is not always better
D. It is guaranteed to be twice as accurate on every task

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q4
Artificial Intelligence, Set 9: Data, patterns and parameters · MAIN · Q4/5 · Outcome AI.LEARN.02 (local)

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Can a model ever reproduce a piece of text it saw during training?

A. Always, because it stores every page exactly
B. Sometimes, especially text it saw many times
C. Only if the user types the password for it
D. Never, because it keeps no trace of any text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-M-Q5
Artificial Intelligence, Set 9: Data, patterns and parameters · MAIN · Q5/5 · Outcome AI.LEARN.02 (local)

Learn
Imagine tuning a guitar by ear. You pluck a string, hear that it is a little flat, and turn the peg slightly. You pluck again and turn again, until it sounds right. You do not write down every sound you heard; what you keep is the final peg position. A model's PARAMETERS are like millions of pegs. Training plucks, listens and turns: each wrong output nudges the pegs a little. What the model keeps afterward is the peg settings, which capture what "sounding right" means for its task.

In the comparison, what are the guitar pegs like?

A. The model's parameters
B. The app's screen
C. The training examples
D. The user's questions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q1
Artificial Intelligence, Set 9: Data, patterns and parameters · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LEARN.02 (local)

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What do you do when a string sounds flat?

A. Stop playing for good
B. Throw the guitar away
C. Write down the sound and keep it forever
D. Turn the peg a little and try again

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q2
Artificial Intelligence, Set 9: Data, patterns and parameters · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LEARN.02 (local)

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After tuning, what do you keep?

A. The broken strings
B. A recording of every note plucked
C. The final peg positions
D. Nothing at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q3
Artificial Intelligence, Set 9: Data, patterns and parameters · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LEARN.02 (local)

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Which part of training is like plucking a string and hearing it is wrong?

A. Seeing that an output does not match an example
B. Deleting the model to start over
C. Buying a faster computer for the model
D. Typing a new question into the finished app

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q4
Artificial Intelligence, Set 9: Data, patterns and parameters · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LEARN.02 (local)

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Why are there "millions of pegs" in a model?

A. It has many users
B. It has very many parameters
C. It runs on many computers
D. It plays very many songs

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-B1-Q5
Artificial Intelligence, Set 9: Data, patterns and parameters · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LEARN.02 (local)

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A news article says a model "contains every book it read." What is more accurate?

A. Models never learn anything from books at all
B. It is accurate; parameters are the books themselves
C. It is accurate; models are libraries of full texts
D. It holds learned settings, not a complete copy of every book

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q1
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LEARN.02 (local)

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Two models: one has 1 billion parameters, the other 10 billion. Which conclusion is justified without testing?

A. Both must give identical answers
B. The smaller one must be useless for every task
C. The larger has more capacity; testing shows which works better
D. The larger one is exactly ten times more accurate on every task

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q2
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LEARN.02 (local)

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A model was trained to recognize cats. What did training adjust?

A. The photos themselves, so they look more like cats
B. Its parameters, so pixel patterns map to "cat"
C. A written rule list describing every cat breed
D. The user's camera settings

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q3
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LEARN.02 (local)

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Why might a model repeat a famous quotation word for word?

A. It looks the quotation up in a stored copy of the internet
B. Models are designed to repeat everything word for word
C. The quotation is one of its parameters
D. Text seen many times in training can leave strong traces

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q4
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LEARN.02 (local)

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Which description of training is most accurate?

A. One big change made after all examples are seen
B. Typing in facts one at a time into a database
C. Many small adjustments to settings, based on errors
D. Copying the examples into the model's memory

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F1-Q5
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LEARN.02 (local)

Learn
Because learned patterns are stored as dial settings rather than as a list of facts, you usually cannot open a model and look up "the" fact it learned or delete one example's influence by hand. This is one reason models can mix patterns in surprising ways: an answer may blend several sources into something no single source said. It also explains why models sometimes reproduce memorized passages: text repeated very often in training can leave strong traces. Claims such as "the model just copies" or "the model never copies" are both too simple.

A user asks a company to "delete the fact about me from the model." Why is this hard?

A. Learned patterns are spread across settings, not stored as one fact
B. Models store each fact in a labeled folder that is locked
C. It is easy; any user can delete one parameter
D. Models never learn anything about people

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q1
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LEARN.02 (local)

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A model's answer blends ideas so that no single source said exactly that. What explains this?

A. It always copies one source exactly
B. It searched one website and pasted it
C. It has a written rule never to use any real sources
D. It combines learned patterns from many examples

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q2
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LEARN.02 (local)

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Someone says, "Models never copy anything, so there is no copyright issue." What is wrong?

A. Nothing; models are proven never to copy at all
B. Models can sometimes reproduce passages they saw many times
C. Models only copy images, never text
D. Copying is impossible because parameters are numbers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q3
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LEARN.02 (local)

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Someone says, "Models just copy and paste their training data." What is wrong?

A. Mostly they produce new combinations from patterns
B. Models only copy text from the user
C. Models cannot produce any text at all
D. Nothing; every output is an exact copy of a source

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q4
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LEARN.02 (local)

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Why can't you open a model and read "the fact" it learned about the moon?

A. Models never learn about the moon
B. The fact is locked away inside it with a secret password
C. It is spread across many numbers, not stored as sentences
D. Models refuse to discuss the moon

REFERENCE LABEL: NOT PART OF THE QUESTION ID S9-F2-Q5
Artificial Intelligence, Set 9: Data, patterns and parameters · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LEARN.02 (local)

Set 10: Training, validation and test data

Outcome AI.LEARN.03: Explain why data is split into training, validation and test sets and why test data must stay unseen during training

Module: Foundation 2: How a model learns

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Learn
Builders usually split their data into three parts. The TRAINING SET is what the model learns from. The VALIDATION SET is used during development to compare versions and settings, like practice quizzes. The TEST SET is kept aside and used only at the end to measure how well the finished model does on examples it has never seen. If test examples leak into training, the test score looks better than the model really is, because the model may have learned those exact answers. A trustworthy result comes from examples that stayed unseen.

What is the training set used for?

A. Advertising the model to customers
B. Comparing versions during development
C. The final check
D. The examples the model learns from

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q1
Artificial Intelligence, Set 10: Training, validation and test data · MAIN · Q1/5 · Outcome AI.LEARN.03 (local)

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What is the test set used for?

A. Tuning settings during development
B. Teaching the model its main patterns
C. Measuring the finished model on unseen examples
D. Storing users' questions after the app launches

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q2
Artificial Intelligence, Set 10: Training, validation and test data · MAIN · Q2/5 · Outcome AI.LEARN.03 (local)

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Which set is most like practice quizzes during development?

A. None of the sets
B. The validation set
C. The test set
D. The training set

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q3
Artificial Intelligence, Set 10: Training, validation and test data · MAIN · Q3/5 · Outcome AI.LEARN.03 (local)

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Why must test examples stay out of training?

A. Otherwise the test score can overstate real performance
B. Otherwise the training will finish too quickly
C. Otherwise the model will refuse to work on any input at all
D. Otherwise the model gets too many parameters

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q4
Artificial Intelligence, Set 10: Training, validation and test data · MAIN · Q4/5 · Outcome AI.LEARN.03 (local)

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A model scores 99 percent, but later it is found that test examples were also in training. What should you conclude?

A. Nothing; leaks never affect scores
B. The 99 percent may overstate how it handles new examples
C. The model must be useless for everything
D. The 99 percent is now even more trustworthy than before

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-M-Q5
Artificial Intelligence, Set 10: Training, validation and test data · MAIN · Q5/5 · Outcome AI.LEARN.03 (local)

Learn
A student preparing for a final exam studies from a TEXTBOOK, takes PRACTICE QUIZZES to see what still needs work, and then takes the FINAL EXAM. The exam is useful only if its questions are new. If the student saw the exact exam questions in advance, a high score would show memory, not real understanding. Model data works the same way: the training set is the textbook, the validation set is the practice quizzes, and the test set is the final exam that must stay secret until the end.

The textbook is like which data set?

A. The training set
B. The validation set
C. The test set
D. No data set

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q1
Artificial Intelligence, Set 10: Training, validation and test data · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LEARN.03 (local)

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The final exam is like which data set?

A. The training set
B. All three sets
C. The validation set
D. The test set

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q2
Artificial Intelligence, Set 10: Training, validation and test data · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LEARN.03 (local)

Learn
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Why must the final exam questions stay secret?

A. So the textbook is not needed
B. So the exam is harder to print
C. So a high score shows understanding
D. So the teacher can grade all the exams faster

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q3
Artificial Intelligence, Set 10: Training, validation and test data · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LEARN.03 (local)

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What do practice quizzes help the student do?

A. Get the final grade early
B. Skip the textbook
C. Memorize the final exam questions early
D. See what still needs work

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q4
Artificial Intelligence, Set 10: Training, validation and test data · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LEARN.03 (local)

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A student saw the exact exam in advance and scored 100. What does the score mostly show?

A. That the exam was too easy
B. Memory of those questions
C. Deep understanding of everything
D. Nothing at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-B1-Q5
Artificial Intelligence, Set 10: Training, validation and test data · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LEARN.03 (local)

Learn
Builders usually split their data into three parts. The TRAINING SET is what the model learns from. The VALIDATION SET is used during development to compare versions and settings, like practice quizzes. The TEST SET is kept aside and used only at the end to measure how well the finished model does on examples it has never seen. If test examples leak into training, the test score looks better than the model really is, because the model may have learned those exact answers. A trustworthy result comes from examples that stayed unseen.

A team building a dog-breed recognizer has 10,000 labeled photos. Which plan follows good practice?

A. Train and test on the same 10,000 photos
B. Test first, then add the test photos into training and report that score
C. Train on most, use some for validation, keep a test portion unseen
D. Skip testing because 10,000 is already enough

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q1
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LEARN.03 (local)

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During development, a team compares three versions and picks the best. Which data should they use to compare?

A. The validation set
B. The training set only
C. No data; they should guess
D. The test set, again and again

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q2
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LEARN.03 (local)

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A student project reports "100 percent accuracy," measured on the same examples it trained on. What is the problem?

A. There is no problem; 100 percent is always trustworthy
B. The score does not show how it handles new examples
C. Training data cannot be used for anything at all
D. Accuracy cannot be measured for student projects

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q3
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LEARN.03 (local)

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When should the test set be used?

A. At the start, to teach the model
B. Every day, to choose settings
C. At the end, to measure the finished model
D. Never, because final tests are always optional

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q4
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LEARN.03 (local)

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A company says its spam filter is "98 percent accurate on our test set." What question checks whether the number is trustworthy?

A. Were the test emails kept out of training?

B. Is 98 a bigger number than last year's?

C. Does the filter have a friendly name?

D. Is the number written in a large font?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F1-Q5
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LEARN.03 (local)

Learn
Keeping test data truly separate is harder than it sounds. Copies of the same item can appear twice, for example the same photo cropped slightly, or the same news story on two websites, so one copy lands in training and another in the test. Reusing the test set again and again to choose between versions also wears it out: the team slowly tunes the model to that particular test. And a test set that does not look like real use, such as clear studio photos when users will upload blurry phone pictures, gives a score that does not predict real performance.

The same news story appears on two websites. One copy is in training, the other in the test set. What is the risk?

A. The model will refuse to read news
B. The test becomes harder than real use
C. None, because the websites have different names
D. The test includes something the model has already seen

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q1
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LEARN.03 (local)

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A team checks its test score after every small change for months. What problem can this cause?

A. The team slowly tunes the model to that one test
B. The model forgets its training data
C. Nothing; checking often always helps
D. The test questions physically wear out from use

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q2
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LEARN.03 (local)

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A plant-identification app was tested on clear studio photos, but users upload blurry phone pictures. What should the team conclude?

A. The test score proves the app works for everyone
B. Blurry photos cannot be identified by anyone
C. The test score may not predict real-world performance
D. The app will work even better on blurry photos

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q3
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LEARN.03 (local)

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Photos of the same dog, cropped slightly differently, end up in both training and test sets. What should the team do?

A. Keep them, because cropping makes them different
B. Remove near-duplicates so test items are truly unseen
C. Add more copies to make the test fairer
D. Delete the whole test set and skip testing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q4
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LEARN.03 (local)

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Which test plan best predicts real performance for a store's receipt scanner?

A. Receipts the scanner was trained on
B. One receipt scanned 100 times
C. Perfect, printed sample receipts from the factory only
D. Unseen receipts like the ones customers really scan

REFERENCE LABEL: NOT PART OF THE QUESTION ID S10-F2-Q5
Artificial Intelligence, Set 10: Training, validation and test data · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LEARN.03 (local)

Set 11: Overfitting and generalization

Outcome AI.LEARN.04: Recognize overfitting (doing well on familiar examples but poorly on new ones) and what good generalization looks like

Module: Foundation 2: How a model learns

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A model GENERALIZES when it does well on new examples, not just the ones it trained on. OVERFITTING is the opposite: the model fits its training examples so closely, including their accidents and noise, that it does worse on new ones. A sign of overfitting is a big gap: very high accuracy on training data but much lower accuracy on validation or test data. For example, a model that learned "all photos of wolves have snow" might label a husky in snow as a wolf. Remedies include more varied data, simpler models and stopping training before it memorizes.

What does it mean for a model to generalize well?

A. It memorizes every one of its training examples exactly
B. It does well on new examples, not just training ones
C. It works only on the examples it trained on
D. It has as many parameters as possible

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q1
Artificial Intelligence, Set 11: Overfitting and generalization · MAIN · Q1/5 · Outcome AI.LEARN.04 (local)

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Which result is a warning sign of overfitting?

A. A model that has not been trained yet
B. 50 percent on training data and 50 percent on test data
C. 99 percent on training data but 70 percent on test data
D. 85 percent on training data and 84 percent on test data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q2
Artificial Intelligence, Set 11: Overfitting and generalization · MAIN · Q2/5 · Outcome AI.LEARN.04 (local)

Learn
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A model learned that wolf photos have snow and calls a husky in snow a wolf. What happened?

A. Huskies and wolves are the same animal
B. It generalized perfectly from good data
C. It was never trained on any photos of animals at all
D. It overfit to an accidental pattern in its photos

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q3
Artificial Intelligence, Set 11: Overfitting and generalization · MAIN · Q3/5 · Outcome AI.LEARN.04 (local)

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Which change could reduce overfitting?

A. Training with more varied examples
B. Testing on the training data instead
C. Removing the validation set
D. Training much longer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q4
Artificial Intelligence, Set 11: Overfitting and generalization · MAIN · Q4/5 · Outcome AI.LEARN.04 (local)

Learn
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Why is overfitting a problem in practice?

A. The training data is deleted automatically
B. The model refuses to accept any input
C. The model becomes too fast to measure
D. The model performs worse on new, real cases

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-M-Q5
Artificial Intelligence, Set 11: Overfitting and generalization · MAIN · Q5/5 · Outcome AI.LEARN.04 (local)

Learn
Two students prepare for a math test. One MEMORIZES the answers to the 20 practice problems: "Problem 4 is 17." The other learns the METHOD for solving them. On the practice sheet, both score perfectly. On the real test, with new numbers, the memorizer is lost while the method-learner does fine. The memorizer has overfit to the practice sheet. The method-learner generalized. A model can do the same: perfect on the examples it trained on, poor on new ones.

Which student overfit?

A. The one who memorized the practice answers
B. Neither student
C. Both students
D. The one who learned the method for solving them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q1
Artificial Intelligence, Set 11: Overfitting and generalization · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LEARN.04 (local)

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Why did both students score perfectly on the practice sheet?

A. Both learned the method
B. Both could answer those exact problems
C. The practice sheet had no problems
D. The teacher gave everyone full marks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q2
Artificial Intelligence, Set 11: Overfitting and generalization · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LEARN.04 (local)

Learn
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What happened on the real test with new numbers?

A. Both failed
B. The memorizer did better than the method-learner
C. The memorizer struggled; the other did fine
D. Both did equally well

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q3
Artificial Intelligence, Set 11: Overfitting and generalization · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LEARN.04 (local)

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Which student is like a model that generalizes?

A. Neither
B. The method-learner
C. Both
D. The memorizing student

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q4
Artificial Intelligence, Set 11: Overfitting and generalization · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LEARN.04 (local)

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What kind of test reveals overfitting?

A. A test the student writes
B. A test with no questions
C. The same practice sheet again
D. A test with new examples

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-B1-Q5
Artificial Intelligence, Set 11: Overfitting and generalization · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LEARN.04 (local)

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A model scores 98 percent on training data and 96 percent on validation data. What is the best reading?

A. It appears to generalize well

B. It is badly overfit

C. It has not learned anything

D. The validation data must have leaked

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q1
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LEARN.04 (local)

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A team's model gets 100 percent in training but 60 percent in validation. What should they try?

A. Stop using any validation data
B. Report the 100 percent as its accuracy
C. More varied data or a simpler model
D. Train longer on the same data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q2
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LEARN.04 (local)

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A spam filter learned that every email containing "meeting" is safe, because all its safe training emails had that word. What could go wrong?

A. Safe emails will all be blocked
B. Nothing; the word "meeting" proves safety
C. The filter will stop reading emails
D. Spam containing "meeting" will slip through

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q3
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Which report best shows a model generalizes?

A. A very large number of parameters
B. Strong results on new examples it never saw
C. Perfect results on its own training examples
D. A long training time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q4
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LEARN.04 (local)

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A photo model always calls cows "horses" when they stand on sand, because training horses were on beaches. What is the fix?

A. Train with animals photographed in varied settings
B. Increase the model's confidence scores for cows
C. Remove all cow photos from the test
D. Show only beach photos from now on

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F1-Q5
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LEARN.04 (local)

Learn
Overfitting is not the only way a model can disappoint. A model can also UNDERFIT: it is too simple or trained too little, so it does poorly even on its training data. And a model can generalize well to data like its training data but still fail when the situation changes, for example a new camera, a new slang word or a new kind of fraud. So compare training and validation results first: a big gap suggests overfitting; low scores on both suggest underfitting; good scores followed by a drop after launch suggest the world has changed.

A model scores 55 percent on training data and 54 percent on validation data for a task where 90 percent is expected. What is the most likely issue?

A. It is working perfectly

B. The world changed after launch

C. Underfitting: it has not learned the task well

D. Overfitting: training is far higher than validation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q1
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LEARN.04 (local)

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A fraud model tested well, but months after launch it misses a new kind of scam. What best explains this?

A. Fraud can never be detected by models
B. The situation changed in ways its training data did not cover
C. It overfit, since training was far above validation
D. It underfit, since it never learned anything

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q2
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LEARN.04 (local)

Learn
Overfitting is not the only way a model can disappoint. A model can also UNDERFIT: it is too simple or trained too little, so it does poorly even on its training data. And a model can generalize well to data like its training data but still fail when the situation changes, for example a new camera, a new slang word or a new kind of fraud. So compare training and validation results first: a big gap suggests overfitting; low scores on both suggest underfitting; good scores followed by a drop after launch suggest the world has changed.

Which result pattern suggests overfitting?

A. High training score, much lower validation score
B. High scores on both, then a drop after launch
C. Low training score, low validation score
D. High scores on both with a small gap

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q3
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LEARN.04 (local)

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A speech model trained on recordings from one microphone does poorly with a new phone model's microphone. What should the team check first?

A. Whether the old test score was rounded
B. Whether the model has too few parameters to handle speech
C. Whether users are speaking too politely
D. Whether the new microphone's audio differs from training data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q4
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LEARN.04 (local)

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A team sees a big training-validation gap and wants to add many more parameters. What is the concern?

A. Validation scores will no longer be measured at all

B. More parameters always fix overfitting

C. A bigger model might memorize more and overfit worse

D. Parameters cannot be added to models

REFERENCE LABEL: NOT PART OF THE QUESTION ID S11-F2-Q5
Artificial Intelligence, Set 11: Overfitting and generalization · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LEARN.04 (local)

Set 12: Data quality and labels

Outcome AI.LEARN.05: Explain how missing, wrong, outdated or unrepresentative data and labels affect what a model learns

Module: Foundation 2: How a model learns

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A model can only learn what its data shows it, so DATA QUALITY matters as much as the model itself. Common problems: MISSING data (some cases never appear), WRONG LABELS (a cat photo labeled "dog"), OUTDATED data (prices from five years ago), and UNREPRESENTATIVE data (photos only from one country, voices only from adults). A LABEL is the correct answer attached to a training example, often added by people, and people make mistakes or disagree. Problems in the data become problems in the model, which is often summed up as "garbage in, garbage out."

A model was trained on product prices from five years ago. Which data problem is this?

A. Wrong labels
B. Outdated data
C. Missing data
D. Too many parameters

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q1
Artificial Intelligence, Set 12: Data quality and labels · MAIN · Q1/5 · Outcome AI.LEARN.05 (local)

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Many cat photos in a training set are labeled "dog." What will most likely happen?

A. The model will recognize cats perfectly anyway
B. The model will refuse to train at all
C. The model may confuse cats and dogs
D. Nothing; labels have no effect on learning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q2
Artificial Intelligence, Set 12: Data quality and labels · MAIN · Q2/5 · Outcome AI.LEARN.05 (local)

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A voice assistant was trained only on adult voices. What problem may appear?

A. It may understand children poorly
B. It will refuse to hear adult voices
C. It will have no problems at all
D. It will understand children better

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q3
Artificial Intelligence, Set 12: Data quality and labels · MAIN · Q3/5 · Outcome AI.LEARN.05 (local)

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What is a label in training data?

A. A sticker on the user's phone
B. The title of the finished app
C. The brand name printed on the computer
D. The correct answer attached to an example

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q4
Artificial Intelligence, Set 12: Data quality and labels · MAIN · Q4/5 · Outcome AI.LEARN.05 (local)

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What does "garbage in, garbage out" mean for AI?

A. Models clean up any bad data automatically
B. Problems in the data become problems in the model
C. Bad data always makes a model refuse to work
D. Only the model's size affects its quality

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-M-Q5
Artificial Intelligence, Set 12: Data quality and labels · MAIN · Q5/5 · Outcome AI.LEARN.05 (local)

Learn
Imagine learning a city from a MAP. If the map is missing a neighborhood, you will never know it exists. If a street is labeled with the wrong name, you will confidently give wrong directions. If the map is twenty years old, you may send people to a store that closed long ago. If the map shows only the downtown, you will be lost in the suburbs. A model learns from its data the way you learn from the map: whatever is missing, wrong, old or one-sided in the data shows up in the model.

A map missing a neighborhood is like which data problem?

A. Unrepresentative data
B. Wrong labels
C. Missing data
D. Outdated data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q1
Artificial Intelligence, Set 12: Data quality and labels · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LEARN.05 (local)

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A street with the wrong name is like which data problem?

A. Wrong labels
B. Unrepresentative data
C. Missing data
D. Outdated data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q2
Artificial Intelligence, Set 12: Data quality and labels · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LEARN.05 (local)

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A twenty-year-old map is like which data problem?

A. Too many streets
B. Missing data
C. Wrong labels
D. Outdated data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q3
Artificial Intelligence, Set 12: Data quality and labels · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LEARN.05 (local)

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A map showing only downtown is like which data problem?

A. Unrepresentative data
B. Wrong labels
C. Outdated data from long ago
D. Perfect data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q4
Artificial Intelligence, Set 12: Data quality and labels · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LEARN.05 (local)

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Why would you give wrong directions confidently with a flawed map?

A. You never look at the map
B. You trust what the map shows you
C. Directions cannot be wrong
D. Maps always tell you

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-B1-Q5
Artificial Intelligence, Set 12: Data quality and labels · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LEARN.05 (local)

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A skin-condition app was trained mostly on photos of light skin. What risk should its makers check?

A. Too many correct answers on light skin
B. The app running too fast on phones
C. Weaker accuracy on darker skin
D. Users taking photos in good lighting

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q1
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LEARN.05 (local)

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Volunteers labeled 10,000 tweets as "angry" or "calm" in a rush and made many mistakes. What is the effect?

A. The model ignores labels and learns correctly
B. The model will be perfect because 10,000 is many
C. Nothing; rushed labels are always fine
D. The model learns some wrong associations

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q2
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LEARN.05 (local)

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A rent-prediction model uses listings from 2019 to predict rents today. Which problem is most likely?

A. Outdated data that misses recent price changes
B. Too much data from today
C. Wrong labels on every listing
D. Perfect predictions because the data is old

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q3
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LEARN.05 (local)

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A traffic model never saw data from snowy days. What could happen on the first snowy day?

A. It will forecast snow by itself
B. Its predictions will be most accurate in snow
C. Its predictions may be badly wrong
D. Nothing will change at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q4
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LEARN.05 (local)

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Which step best improves the quality of a training set?

A. Using only the newest computer
B. Checking a sample of labels for mistakes
C. Hiding where all of the data came from
D. Making the file names shorter

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F1-Q5
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LEARN.05 (local)

Learn
Fixing data problems takes judgment. Some labels are hard because people genuinely disagree: is a review that says "not bad" positive or neutral? Writing clear labeling instructions and checking a sample of labels helps. Collecting more data does not help if the new data has the same gaps. And removing a column such as age does not always remove its influence, because other data can stand in for it. Good teams document where their data came from, what it leaves out and how labels were checked.

Labelers disagree whether "not bad" is positive or neutral. What is the best step?

A. Delete every review that is hard to label
B. Label every review "positive" to be safe
C. Let each labeler decide freely with no guidance
D. Write clear labeling instructions and check a sample

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q1
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LEARN.05 (local)

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A team adds 50,000 more photos, but all from the one city already in the data. Why might this not help?

A. Photos from one city cannot be used at all

B. More data always fixes every problem

C. The new data has the same gaps as the old data

D. The model will forget the old photos

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q2
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LEARN.05 (local)

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A loan model has no "age" column, but it uses "years since first credit card." Why might age still affect it?

A. Other data can stand in for the removed column
B. Credit cards have nothing to do with age
C. Models cannot use numbers about time
D. Removing a column always removes its influence

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q3
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Why should a team document where its data came from and what it leaves out?

A. So the model trains faster
B. So that no one asks any questions about it
C. So the data file looks longer
D. So others can judge where the model may fail

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q4
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LEARN.05 (local)

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A new labeling team is much faster but less careful. What should the project do?

A. Hide their labels from the model
B. Check a sample of their labels before trusting them
C. Trust them, because working fast shows real skill
D. Stop labeling data forever

REFERENCE LABEL: NOT PART OF THE QUESTION ID S12-F2-Q5
Artificial Intelligence, Set 12: Data quality and labels · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LEARN.05 (local)

Set 13: Fine-tuning versus giving information in a request

Outcome AI.LEARN.08: Tell fine-tuning (further training that changes a model) from supplying information in a request (which does not change the model)

Module: Foundation 2: How a model learns

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
There are two very different ways to make a model handle your needs. FINE-TUNING is more training: a model that already exists is trained further on new examples, which changes its parameters. It takes data, time and skill, and the change lasts for every later use of that tuned version. SUPPLYING INFORMATION IN THE REQUEST means putting what the model needs, such as a policy document, a list of rules or a few examples, into your message or an attached file. That does not change the model; it only affects that request (and that conversation, while the text stays in context).

A company trains an existing model further on 20,000 of its own support chats. What is this?

A. Deleting the model's parameters
B. Inference on 20,000 new questions
C. Fine-tuning, which changes the model
D. Supplying information in a request

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q1
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · MAIN · Q1/5 · Outcome AI.LEARN.08 (local)

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Leo pastes his school's rules into his message and asks a question about them. What happens to the model?

A. It permanently forgets other schools' rules
B. Nothing changes; the rules affect only this request
C. It shares the rules with every other user
D. It is fine-tuned on the school rules forever

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q2
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · MAIN · Q2/5 · Outcome AI.LEARN.08 (local)

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Which approach changes the model's parameters?

A. Typing a longer, clearer question
B. Attaching a file to one message
C. Adding three examples to a request
D. Fine-tuning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q3
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · MAIN · Q3/5 · Outcome AI.LEARN.08 (local)

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What does fine-tuning usually require?

A. Example data, time and some skill
B. Only one sentence typed into a chat
C. Deleting the original model first
D. Nothing; it happens automatically

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q4
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · MAIN · Q4/5 · Outcome AI.LEARN.08 (local)

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How long does information supplied in a request affect the model's answers?

A. Exactly one week, then it expires
B. Only while it is in that request or conversation
C. Until the company retrains the model
D. Forever, for every user of the model everywhere

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-M-Q5
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · MAIN · Q5/5 · Outcome AI.LEARN.08 (local)

Learn
A new employee can learn about a company in two ways. She can attend a TRAINING COURSE for weeks, and what she learns stays with her afterward. Or her manager can leave a NOTE on her desk: "Today the store closes at 6." The note helps today, but it does not change her training, and tomorrow there may be a different note. Fine-tuning is like the training course: it changes the model. Putting information in a request is like the note: it helps with that task and changes nothing permanent.

Fine-tuning is like which option?

A. Both options
B. The note left on the desk
C. The weeks-long training course
D. Neither option

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q1
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LEARN.08 (local)

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Pasting information into a request is like which option?

A. The note on the desk
B. Hiring a new employee
C. The training course
D. Closing the store

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q2
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LEARN.08 (local)

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Does the note change what the employee learned in training?

A. No, because she cannot read

B. Yes, it replaces the course

C. Yes

D. No, it only helps today

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Which lasts after the task is done?

A. What she learned in the course
B. Neither
C. The note's message
D. Only the store hours in the note

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q4
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LEARN.08 (local)

Learn
A new employee can learn about a company in two ways. She can attend a TRAINING COURSE for weeks, and what she learns stays with her afterward. Or her manager can leave a NOTE on her desk: "Today the store closes at 6." The note helps today, but it does not change her training, and tomorrow there may be a different note. Fine-tuning is like the training course: it changes the model. Putting information in a request is like the note: it helps with that task and changes nothing permanent.

Tomorrow the store closes at 8. What is the easiest way to tell her?

A. Tell no one
B. Hire a different employee
C. Leave a new note
D. Send her back to a weeks-long course

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-B1-Q5
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LEARN.08 (local)

Learn
There are two very different ways to make a model handle your needs. FINE-TUNING is more training: a model that already exists is trained further on new examples, which changes its parameters. It takes data, time and skill, and the change lasts for every later use of that tuned version. SUPPLYING INFORMATION IN THE REQUEST means putting what the model needs, such as a policy document, a list of rules or a few examples, into your message or an attached file. That does not change the model; it only affects that request (and that conversation, while the text stays in context).

A small shop wants answers about its return policy, which it updates often. Which approach is simplest to keep current?

A. Hope the model already knows the shop's policy
B. Supply the current policy in each request
C. Fine-tune a model every time the policy changes
D. Fine-tune once and never update it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q1
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LEARN.08 (local)

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Tia attached a recipe file and asked for a shopping list. Did the model learn her recipe permanently?

A. Yes; attaching a file fine-tunes the model

B. Yes; all users can now see her recipe

C. No; models cannot read attached files

D. No; the file was used only for that request

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q2
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LEARN.08 (local)

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A law firm wants a model to always write in its strict house format across thousands of documents. Which approach might suit it?

A. Never giving any examples
B. Typing "use our format" once and assuming it lasts forever
C. Fine-tuning on many examples of that format
D. Asking a different model each time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q3
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LEARN.08 (local)

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Which statement about fine-tuning is accurate?

A. It changes the model for later uses of that version
B. It needs no data at all
C. It happens whenever a user says "learn this"
D. It changes only the current message

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q4
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LEARN.08 (local)

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A teacher wants a model to answer questions about this week's homework sheet. What is the easiest approach?

A. Ask the model to guess the sheet
B. Expect the model to know the sheet already
C. Fine-tune a model on the sheet
D. Include the homework sheet in the request

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F1-Q5
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LEARN.08 (local)

Learn
Choosing between the two depends on the job. For facts that change often, like this week's prices or a policy updated monthly, supplying current information in the request is usually easier to keep up to date. Fine-tuning suits teaching a lasting style or format across many uses, or a specialized task with lots of examples. Fine-tuning is NOT a reliable way to make a model "know" a long list of changing facts, and it can introduce new mistakes. Either way, results need testing, and private data used for fine-tuning needs the same care as any other data sharing.

A store wants a model to know its 2,000 prices, which change weekly. Why is fine-tuning a poor fit?

A. Prices are illegal to share with any AI model
B. Prices change weekly; fine-tuning suits that poorly
C. Fine-tuning works only for images
D. Fine-tuning cannot use numbers at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q1
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LEARN.08 (local)

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After fine-tuning, a model starts making errors it did not make before. What does this show?

A. Fine-tuning can add new mistakes, so test the result
B. The original model was deleted by accident
C. Fine-tuning always makes every answer better
D. Errors cannot happen after fine-tuning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q2
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LEARN.08 (local)

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A team wants to fine-tune on customer emails that include home addresses. What should they consider first?

A. Nothing; private data is fine to use for any kind of training
B. Whether they may use that private data, and how to protect it
C. Only how fast the training will finish
D. Whether the emails are long enough

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q3
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LEARN.08 (local)

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Which task is the best fit for supplying information in the request rather than fine-tuning?

A. Creating a new version for long-term use
B. Training a model on a large specialized task
C. Answering questions about this month's updated policy
D. Teaching a lasting writing style used across all documents

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q4
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LEARN.08 (local)

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A manager says, "We fine-tuned it, so we no longer need to check its answers." What is wrong?

A. Fine-tuning deletes the ability to check
B. Checking is only for models that were never tuned
C. Nothing; fine-tuned models are always correct
D. Tuned models still make mistakes and need testing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S13-F2-Q5
Artificial Intelligence, Set 13: Fine-tuning versus giving information in a request · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LEARN.08 (local)

Set 14: What a large language model is

Outcome AI.LLM.01: Describe a large language model (LLM) as a model trained on large amounts of text to predict likely next pieces of text, and say what that does and does not imply

Module: Foundation 3: Large language models

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A LARGE LANGUAGE MODEL (LLM) is a model trained on enormous amounts of text to do one core job: given some text, predict a likely next piece of text. Repeating that prediction, piece by piece, produces sentences, answers, stories or code. Because it learned from so much writing, an LLM can imitate many styles and often produces useful, relevant text. But predicting likely text is not the same as looking up facts or reasoning like a person. An LLM has no built-in way to check truth, so its output can be plausible and wrong. "Large" refers to the size of the model and its training data.

What is the core job a large language model is trained to do?

A. Store a copy of the internet to read from
B. Predict a likely next piece of text
C. Look up each answer in a database of verified facts
D. Follow a fixed rule list written by programmers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q1
Artificial Intelligence, Set 14: What a large language model is · MAIN · Q1/5 · Outcome AI.LLM.01 (local)

Learn
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What does "large" refer to in "large language model"?

A. The number of languages printed on its box
B. The length of the answers it is allowed to write
C. The size of the model and its training data
D. The physical size of the computer screen it uses

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q2
Artificial Intelligence, Set 14: What a large language model is · MAIN · Q2/5 · Outcome AI.LLM.01 (local)

Learn
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How does an LLM produce a whole paragraph?

A. By repeating next-piece predictions one after another
B. By copying one paragraph from a single website
C. By asking a human writer to type each sentence
D. By choosing a finished paragraph from a stored list

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q3
Artificial Intelligence, Set 14: What a large language model is · MAIN · Q3/5 · Outcome AI.LLM.01 (local)

Learn
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Why can an LLM's answer be plausible but wrong?

A. It is programmed to make mistakes on purpose
B. It writes only random words with no pattern
C. It always checks facts before it writes
D. Likely-sounding text is not always true

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q4
Artificial Intelligence, Set 14: What a large language model is · MAIN · Q4/5 · Outcome AI.LLM.01 (local)

Learn
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Which task is an LLM usually well suited to help with?

A. Certifying that a medicine dose is safe for you
B. Proving a fact is true just by stating it
C. Drafting a first version of a short letter
D. Guaranteeing the exact current price of a stock

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-M-Q5
Artificial Intelligence, Set 14: What a large language model is · MAIN · Q5/5 · Outcome AI.LLM.01 (local)

Learn
When you type a message, your phone may suggest the next word: after "See you," it might offer "soon" or "tomorrow." It suggests words that often come next, not words it has checked are true for you. A large language model works on a similar core idea at a far bigger scale. It was trained on huge amounts of text, so it can continue whole sentences and paragraphs in many styles. The scale makes it much more capable than a phone keyboard, but the basic job is still predicting likely text.

After "See you," why might a phone suggest "soon"?

A. It knows when you will meet
B. "Soon" often comes next in messages
C. It was told by your friend
D. It checked your calendar for the date

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q1
Artificial Intelligence, Set 14: What a large language model is · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LLM.01 (local)

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What core job do the phone keyboard and an LLM share?

A. Taking photos of the screen
B. Sending messages for you
C. Checking facts against a database
D. Predicting likely next text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q2
Artificial Intelligence, Set 14: What a large language model is · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LLM.01 (local)

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What makes an LLM much more capable than a phone keyboard?

A. Its far bigger scale of training
B. It only works offline
C. It uses fewer words
D. It has a much larger screen

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q3
Artificial Intelligence, Set 14: What a large language model is · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LLM.01 (local)

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Does a likely next word have to be true for you?

A. No, because phones cannot spell

B. Yes, likely words are always true

C. No, likely is not the same as true

D. Yes, if the phone is expensive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q4

Artificial Intelligence, Set 14: What a large language model is · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LLM.01 (local)

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Which continuation would a keyboard most likely suggest after "Happy"?

A. "birthday"
B. "tax refund deadline"
C. "spreadsheet"
D. "carburetor"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-B1-Q5
Artificial Intelligence, Set 14: What a large language model is · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LLM.01 (local)

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Rosa asks an LLM-based chatbot for the population of her town and gets a precise number. What should she do before using it in a report?

A. Round it up so it looks more official

B. Check the number in an official source

C. Ask the chatbot to repeat the same number

D. Use it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q1
Artificial Intelligence, Set 14: What a large language model is · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LLM.01 (local)

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A student wants help turning bullet notes into a clear paragraph. Is an LLM a reasonable tool to try?

A. No; LLMs cannot produce paragraphs
B. Yes; and the result needs no checking
C. No; only search engines can rewrite notes
D. Yes; rewriting suits it, then check the facts

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q2
Artificial Intelligence, Set 14: What a large language model is · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LLM.01 (local)

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Which description of an LLM is most accurate for a beginner?

A. A search engine that only shows links
B. A model that writes by predicting text, which can be wrong
C. A robot that understands everything just like a person
D. A talking encyclopedia that is always correct

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q3
Artificial Intelligence, Set 14: What a large language model is · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LLM.01 (local)

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An LLM writes a confident paragraph about a law. What is the right attitude?

A. Ignore it completely, since LLMs never help
B. Trust it, because legal wording sounds official
C. Treat it as a draft and check the law's actual text
D. Assume it is current because it sounds confident

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q4
Artificial Intelligence, Set 14: What a large language model is · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.01 (local)

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Why might two well-written LLM answers about the same history event disagree?

A. Each is predicted text; either could contain errors
B. History has no facts that can be checked
C. LLMs always agree on facts
D. One of them must have been copied from a textbook

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F1-Q5
Artificial Intelligence, Set 14: What a large language model is · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.01 (local)

Learn
People sometimes describe LLMs with extreme claims: "it is just autocomplete, so it can do nothing useful," or "it knows everything, so it is always right." Both mislead. Predicting text well, at large scale, produces abilities that are genuinely useful, such as drafting, summarizing and explaining. Yet the same process can produce confident errors, because a likely-sounding sentence and a true sentence are not always the same. Judge an LLM by testing it on your task and checking facts, not by a slogan about what it "really" is.

A friend says, "LLMs are just autocomplete, so they are useless." What is the best response?

A. Correct; they cannot help with any real task
B. Correct; autocomplete never produces sentences
C. Wrong; LLMs never predict text at all
D. Prediction at large scale can still be very useful

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q1
Artificial Intelligence, Set 14: What a large language model is · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.01 (local)

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A coworker says, "It knows everything, so I never check it." What is the risk?

A. Confident but wrong answers will slip through
B. The model will stop working if not checked
C. Checking is illegal for business tools
D. None; LLMs cannot make errors

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q2
Artificial Intelligence, Set 14: What a large language model is · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.01 (local)

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What is the best way to judge whether an LLM is good for your task?

A. Count how many people use it worldwide
B. Test it on examples of your task and check the facts
C. Ask it whether it is good at your task
D. Decide based on a slogan about what it "really" is

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q3
Artificial Intelligence, Set 14: What a large language model is · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LLM.01 (local)

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An LLM summarized a report well but added a statistic not in the report. What does this show?

A. It is always right about statistics
B. Summaries never contain errors
C. It is useless for every kind of task at all
D. It can be useful and still make confident errors

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q4
Artificial Intelligence, Set 14: What a large language model is · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LLM.01 (local)

Learn
People sometimes describe LLMs with extreme claims: "it is just autocomplete, so it can do nothing useful," or "it knows everything, so it is always right." Both mislead. Predicting text well, at large scale, produces abilities that are genuinely useful, such as drafting, summarizing and explaining. Yet the same process can produce confident errors, because a likely-sounding sentence and a true sentence are not always the same. Judge an LLM by testing it on your task and checking facts, not by a slogan about what it "really" is.

Why can a likely-sounding sentence be false?

A. Likely sentences are always checked against sources first
B. LLMs cannot write sentences
C. Prediction favors typical-sounding text, not checked text
D. False sentences are always unlikely

REFERENCE LABEL: NOT PART OF THE QUESTION ID S14-F2-Q5
Artificial Intelligence, Set 14: What a large language model is · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.01 (local)

Set 15: Tokens

Outcome AI.LLM.02: Explain tokens as the pieces of text a language model reads and writes, and why token counts differ from word counts

Module: Foundation 3: Large language models

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Language models do not read words the way people do. They read and write TOKENS: chunks of text that may be a whole word, part of a word, a punctuation mark or a space. A common word like "the" is often one token, while a long or rare word like "unbelievably" may be split into several. For English, one provider's rule of thumb is about 4 characters, or about three quarters of a word, per token, but the exact count depends on the model. Token counts matter because limits on how much text a model can handle, and many usage prices, are measured in tokens, not words.

What is a token?

A. A chunk of text, such as a word or part of one
B. A coin you pay with to use an AI website
C. A password that unlocks a model
D. A whole page of text read at once

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q1
Artificial Intelligence, Set 15: Tokens · MAIN · Q1/5 · Outcome AI.LLM.02 (local)

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Which word is (a very common short word) most likely split into several tokens?

A. the (a very common short word)
B. and (a very common short word)
C. unbelievably (a long, rarer word)
D. is (a very common short word)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q2
Artificial Intelligence, Set 15: Tokens · MAIN · Q2/5 · Outcome AI.LLM.02 (local)

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Using the rule of thumb of about three quarters of a word per token, about how many tokens is a 300-word English text?

A. About 225 tokens
B. About 400 tokens
C. About 1,200 tokens
D. About 300 tokens

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q3
Artificial Intelligence, Set 15: Tokens · MAIN · Q3/5 · Outcome AI.LLM.02 (local)

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Why do token counts matter to users?

A. Tokens decide whether the answer is true
B. Tokens are only used for counting images
C. Tokens decide the font the answer appears in
D. Many limits and prices are measured in tokens

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q4
Artificial Intelligence, Set 15: Tokens · MAIN · Q4/5 · Outcome AI.LLM.02 (local)

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Is the token rule of thumb exact for every model?

A. No; counts depend on how each model splits text
B. No; tokens are always exactly one letter
C. Yes; one token always equals one word
D. Yes; every model splits text identically

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-M-Q5
Artificial Intelligence, Set 15: Tokens · MAIN · Q5/5 · Outcome AI.LLM.02 (local)

Learn
A simple estimate turns words into tokens. One provider's rule of thumb for English is that a token is about three quarters of a word. To estimate tokens from words, DIVIDE the number of words by 0.75, which is the same as multiplying by 4 and then dividing by 3. Example: 60 words / 0.75 = 80 tokens. To estimate words from tokens, MULTIPLY tokens by 0.75. Example: 200 tokens x 0.75 = 150 words. These are rough estimates for ordinary English, not exact counts.

About how many tokens is 30 words?

A. 22.5
B. 40
C. 120
D. 30

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q1
Artificial Intelligence, Set 15: Tokens · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.LLM.02 (local)

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About how many words is 400 tokens?

A. 400
B. 533
C. 300
D. 100

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q2
Artificial Intelligence, Set 15: Tokens · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.LLM.02 (local)

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About how many tokens is 150 words?

A. 112.5
B. 150
C. 600
D. 200

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Dividing by 0.75 gives the same result as which steps?

A. Multiply by 4, then divide by 3
B. Add 75 to the number
C. Subtract a quarter
D. Multiply by 3, then divide by 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-B1-Q4
Artificial Intelligence, Set 15: Tokens · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.LLM.02 (local)

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About how many words is 1,000 tokens?

A. 250
B. 1,333
C. 1,000
D. 750

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A tool allows about 2,000 tokens per message. About how many words of English is that?

A. About 2,667 words
B. About 500 words
C. About 1,500 words
D. Exactly 2,000 words

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q1
Artificial Intelligence, Set 15: Tokens · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LLM.02 (local)

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Jun's essay is 900 words. About how many tokens will it use?

A. Exactly 900 tokens
B. About 1,200 tokens
C. About 3,600 tokens
D. About 675 tokens

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q2
Artificial Intelligence, Set 15: Tokens · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LLM.02 (local)

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Why might a message hit a token limit even though it seems short in words?

A. Some words and symbols use several tokens each
B. Token limits only count pictures
C. Every word always uses exactly ten tokens
D. Spaces are never counted at all

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A pricing page charges by tokens. What does that mean for a long document?

A. Long documents are always free
B. Cost depends on the font size used
C. Longer text usually costs more tokens
D. Cost depends only on the number of pages printed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q4
Artificial Intelligence, Set 15: Tokens · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.02 (local)

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Two tools report different token counts for the same paragraph. What is the likely reason?

A. One tool must be broken
B. Tokens are random numbers
C. Paragraphs have no fixed text inside them
D. They split text into tokens differently

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F1-Q5
Artificial Intelligence, Set 15: Tokens · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.02 (local)

Learn
Token counts can surprise you. Text in some languages, unusual spellings, long numbers, code and tables often use more tokens per word than everyday English, so the same message length in words can cost more tokens. Because a model sees tokens rather than letters, tasks such as counting the letters in a word or reversing a word can trip it up even when it writes fluent sentences. And because different models split text differently, a count from one tool is only an estimate for another.

A message with a long table of numbers uses many more tokens than a story of the same word count. Why?

A. Tables are always converted into images first
B. Numbers and table symbols often take more tokens
C. Numbers always count as zero tokens
D. Stories are never counted in tokens

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q1
Artificial Intelligence, Set 15: Tokens · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.02 (local)

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A fluent model miscounts how many times the letter r appears in "strawberry." What helps explain this?

A. It cannot read English at all
B. Letter counting is always easy for every model
C. It works with tokens, not individual letters
D. The word is spelled differently for computers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q2
Artificial Intelligence, Set 15: Tokens · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.02 (local)

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You estimate a cost using one tool's token counter, but you will use a different model. What should you remember?

A. The count is only an estimate for the other model

B. Token counts never affect cost

C. You must count tokens by hand instead

D. The count will match exactly for every model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q3
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A message in a language other than English uses more tokens than its English translation. What is the best explanation?

A. Tokens count only English letters
B. Languages other than English are not allowed to be tokenized
C. The translation must have deleted words
D. Some languages use more tokens per word for some models

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q4
Artificial Intelligence, Set 15: Tokens · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LLM.02 (local)

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A code snippet of 50 lines seems short but uses a lot of tokens. What is the likely reason?

A. Each line of code always counts as exactly one token
B. Symbols, spaces and names in code split into many tokens
C. Code is turned into pictures first
D. Code is never counted in tokens

REFERENCE LABEL: NOT PART OF THE QUESTION ID S15-F2-Q5
Artificial Intelligence, Set 15: Tokens · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.02 (local)

Set 16: Probabilities and why answers vary

Outcome AI.LLM.03: Explain that a language model chooses among likely next tokens, so the same request can produce different answers

Module: Foundation 3: Large language models

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Learn
At each step, a language model gives every possible next token a PROBABILITY: how likely it is to come next. After "The sky is," the token "blue" might get a high probability and "green" a low one. The model then picks a token, often by SAMPLING, which means choosing at random but weighted by those probabilities. That is why the same request can produce different answers on different tries. Some products let users or developers adjust settings, sometimes called temperature, that make choices more predictable or more varied. Variation is useful for brainstorming but means you should not expect identical answers every time.

What does a language model assign to each possible next token?

A. A probability of how likely it is to come next
B. A price for using that particular token
C. A fixed rule saying which token is required
D. A grade showing whether it is true or false

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q1
Artificial Intelligence, Set 16: Probabilities and why answers vary · MAIN · Q1/5 · Outcome AI.LLM.03 (local)

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Prepared example (made-up probabilities for teaching)

Possible next token after "The sky is"	Probability (made-up numbers)
blue	0.62
clear	0.18
gray	0.12
green	0.01
all other tokens	0.07

Using the table, which token is the most likely next choice?

A. clear

B. gray

C. green

D. blue

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q2

Artificial Intelligence, Set 16: Probabilities and why answers vary · MAIN · Q2/5 · Outcome AI.LLM.03 (local)

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blue	0.62
clear	0.18
gray	0.12
green	0.01
all other tokens	0.07

Using the table, what is the combined probability of "clear" or "gray"?

A. 0.80

B. 0.30

C. 0.18

D. 0.12

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q3

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Why can the same request produce different answers on different tries?

A. Someone edits the answers by hand each time
B. The internet changes the model every second
C. The model often samples among likely tokens
D. The model forgets its training each time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q4
Artificial Intelligence, Set 16: Probabilities and why answers vary · MAIN · Q4/5 · Outcome AI.LLM.03 (local)

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Settings such as "temperature" usually change what?

A. Whether the facts are true
B. How warm the computer runs
C. Which language the user speaks and reads
D. How predictable or varied the choices are

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-M-Q5
Artificial Intelligence, Set 16: Probabilities and why answers vary · MAIN · Q5/5 · Outcome AI.LLM.03 (local)

Learn
A PROBABILITY is a number from 0 to 1 that says how likely something is: 0 means impossible, 1 means certain, 0.5 means as likely as not. Probabilities for all the options in one choice add up to 1. To find the chance of "this OR that" when they cannot both happen, ADD their probabilities: if the chance of rain is 0.3 and of snow is 0.1, the chance of rain or snow is 0.4. A higher number means more likely, but a likely option is not guaranteed to happen.

Which probability means "certain"?

A. 0.5
B. 1
C. 0.1
D. 0

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q1
Artificial Intelligence, Set 16: Probabilities and why answers vary · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.LLM.03 (local)

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Rain 0.3, snow 0.1. What is the chance of rain or snow?

A. 0.2
B. 0.03
C. 0.4
D. 0.3

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q2
Artificial Intelligence, Set 16: Probabilities and why answers vary · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.LLM.03 (local)

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Three options have probabilities 0.5, 0.3 and a missing value. What is the missing value?

A. 0.2
B. 1.0
C. 0.8
D. 0.5

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Which option is most likely: A 0.25, B 0.45, C 0.30?

A. A
B. B
C. All are equal
D. C

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q4
Artificial Intelligence, Set 16: Probabilities and why answers vary · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.LLM.03 (local)

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An option has probability 0.9. Is it guaranteed to happen?

A. No, it is likely but not certain
B. Yes, 0.9 equals 1
C. No, it is impossible
D. Yes, anything over 0.5 is certain

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-B1-Q5
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Ali asks for a party theme twice and gets two different ideas. What does this most likely mean?

A. Only one of the ideas could be a real party theme
B. His first request was deleted by mistake
C. The model is broken and should be reported
D. The model sampled differently; both are normal

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q1
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LLM.03 (local)

Learn
At each step, a language model gives every possible next token a PROBABILITY: how likely it is to come next. After "The sky is," the token "blue" might get a high probability and "green" a low one. The model then picks a token, often by SAMPLING, which means choosing at random but weighted by those probabilities. That is why the same request can produce different answers on different tries. Some products let users or developers adjust settings, sometimes called temperature, that make choices more predictable or more varied. Variation is useful for brainstorming but means you should not expect identical answers every time.

A teacher wants many different practice sentences. What does variation offer?

A. A guarantee that every sentence is correct
B. The same sentence every single time
C. Several different options from similar requests
D. Nothing useful for this task

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q2
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LLM.03 (local)

Learn
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A tool offers a setting for "more creative" or "more focused" answers. Which choice fits writing a list of unusual story ideas?

A. More creative (more varied)
B. More focused (more predictable)
C. It makes no difference at all
D. Turn the tool off

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q3
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LLM.03 (local)

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Why is a likely next token not guaranteed to be correct?

A. Likely tokens are always true facts
B. Likely means common in patterns, not checked as true
C. Correct tokens always get a probability of exactly 1
D. Probabilities are only used for pictures

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q4
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.03 (local)

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A user wants three different subject lines for an email. What is the best request?

A. Ask for exactly one line and copy it three times
B. Never use AI for subject lines
C. Ask for three distinct options in one request
D. Ask once and expect the same line every time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F1-Q5
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.03 (local)

Learn
Variation has practical effects. If you ask the same factual question twice and get two different numbers, at most one can be right, and maybe neither, so the difference is a signal to check a source, not to pick the one you like. When you need consistency, such as the same format for 100 product descriptions, give clear instructions and a template, and check the results, because even predictable settings do not guarantee identical outputs in every product. When you want ideas, variation is a feature: ask for several options at once.

You ask a model the height of a famous tower twice and get 324 meters and 330 meters. What should you do?

A. Ask a third time and trust whichever number repeats

B. Average them to 327 and report that

C. Pick the number you like better

D. Check a reliable source; at most one is right

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q1
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.03 (local)

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A store needs 100 product descriptions in exactly the same format. What helps most?

A. Assuming outputs are always identical
B. Clear instructions, a template, and checking the results
C. Higher variation settings for more creativity
D. Asking without any format and hoping it matches

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q2
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.03 (local)

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Even with "predictable" settings, a product gives slightly different wording each time. What is the best conclusion?

A. The answers are therefore all false
B. The settings must be ignored completely by every product
C. The model is learning from each request
D. Settings reduce variation but may not guarantee identical output

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q3
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LLM.03 (local)

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When is variation in answers most helpful?

A. When you want many ideas to choose from
B. When you need a precise measurement
C. When you need one exact legal date
D. When you must match a strict template

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q4
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LLM.03 (local)

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Two runs of the same request give different conclusions about whether a contract clause is risky. What should a careful user do?

A. Run it ten times and take the most common answer
B. Ignore the clause entirely
C. Treat it as uncertain and get the clause reviewed
D. Choose the more reassuring conclusion

REFERENCE LABEL: NOT PART OF THE QUESTION ID S16-F2-Q5
Artificial Intelligence, Set 16: Probabilities and why answers vary · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.03 (local)

Set 17: Fluent is not the same as true

Outcome AI.LLM.04: Explain why smooth, confident language is not evidence that a statement is correct

Module: Foundation 3: Large language models

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Language models are trained to produce text that reads naturally, so their answers are usually FLUENT: smooth, confident and well organized. Fluency is not EVIDENCE. A model can describe a book that does not exist, invent a statistic, or misstate a rule in the same confident tone it uses for correct facts. Signs such as precise numbers, formal language or phrases like "studies show" do not prove anything by themselves. To judge a claim, look for something you can check: a source you can open, a calculation you can redo, or a fact you can confirm elsewhere.

Why do language model answers usually sound smooth and confident?

A. They have checked every fact before writing
B. They copy only from verified textbooks
C. They are always written by human experts
D. They are trained to produce natural-sounding text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q1
Artificial Intelligence, Set 17: Fluent is not the same as true · MAIN · Q1/5 · Outcome AI.LLM.04 (local)

Learn

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Fictional example (prepared in advance; not a real AI output)

Request:

Who wrote the novel "The Silver Lantern of Brightwater"?

Response:

The novel "The Silver Lantern of Brightwater" was written by Eleanor Marsh and published in 1962. It won praise for its vivid portrait of a fishing town and is often taught in literature classes.

In this fictional example, the book title was invented for the question. What does the response show?

A. A confident, detailed answer about something unreal

B. That the model checked a library catalog

C. That any answer with a year must be correct

D. Proof that the book exists, since details are given

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q2

Artificial Intelligence, Set 17: Fluent is not the same as true · MAIN · Q2/5 · Outcome AI.LLM.04 (local)

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An answer says, "Studies show 73 percent of people prefer this." What does that sentence prove by itself?

A. That a real study found exactly 73 percent
B. That the number is rounded correctly
C. Nothing yet; you would need the actual studies
D. That most people prefer it, for certain

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q3
Artificial Intelligence, Set 17: Fluent is not the same as true · MAIN · Q3/5 · Outcome AI.LLM.04 (local)

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Which of these is something you can actually check?

A. A very precise-looking number with decimals
B. A link to the report that you can open
C. A confident, formal tone
D. The phrase "experts agree"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q4
Artificial Intelligence, Set 17: Fluent is not the same as true · MAIN · Q4/5 · Outcome AI.LLM.04 (local)

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A model gives a correct-sounding rule for a game. What is the best way to confirm it?

A. Ask the model whether it is sure
B. Trust it because it is well organized
C. Assume it is right since it is fluent
D. Check the official rulebook

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-M-Q5
Artificial Intelligence, Set 17: Fluent is not the same as true · MAIN · Q5/5 · Outcome AI.LLM.04 (local)

Learn
Two questions help with any claim. First: HOW DOES IT SOUND? Confident, formal, detailed, friendly. Second: WHAT SUPPORTS IT? A source you can open, a calculation you can redo, a fact you can confirm elsewhere. Only the second question tells you whether to trust it. Example: "The museum opens at 9 a.m., according to its website (link)" has support you can check. "The museum definitely opens at 9 a.m." only sounds sure. A claim can sound weak and be true, or sound strong and be false.

Which part tells you whether to trust a claim?

A. How long it is
B. What supports it
C. How formal the words are
D. How confident it sounds

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q1
Artificial Intelligence, Set 17: Fluent is not the same as true · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.LLM.04 (local)

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Which claim includes support you can check?

A. "Everyone knows it opens at 9"
B. "Definitely opens at 9 a.m."
C. "Opens at 9 a.m., per its website (link)"
D. "It surely opens at 9 a.m., just trust me"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q2
Artificial Intelligence, Set 17: Fluent is not the same as true · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.LLM.04 (local)

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"The bridge is 2.3 km long, I am completely certain." What kind of statement is "I am completely certain"?

A. How it sounds, not support
B. A source that you can open
C. An official record
D. A calculation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q3
Artificial Intelligence, Set 17: Fluent is not the same as true · BACKWARD 1 (underlying skill) · Q3/5 · Outcome AI.LLM.04 (local)

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Can a claim sound unsure and still be true?

A. No; true claims always sound sure
B. Only if it is very long
C. Yes; how it sounds does not decide truth
D. No; unsure claims are always false

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q4
Artificial Intelligence, Set 17: Fluent is not the same as true · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.LLM.04 (local)

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Which is a calculation you can redo?

A. "The price is surely fair"
B. "Everyone online says the tickets are cheap"
C. "The tickets are a great deal"
D. "3 tickets at 12 dollars each is 36 dollars"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-B1-Q5
Artificial Intelligence, Set 17: Fluent is not the same as true · BACKWARD 1 (underlying skill) · Q5/5 · Outcome AI.LLM.04 (local)

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A chatbot gives a detailed biography of a local artist, including awards. Nothing else online mentions them. What is the most reasonable conclusion?

A. Some details may be invented; confirm them elsewhere
B. Search engines must be hiding the awards
C. The biography must be true because it is detailed
D. The artist must be famous but very private

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q1
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Sara needs a statistic for a speech. The model gives one with no source. What should she do?

A. Use it but round it to sound safer
B. Find the original source before using it
C. Ask the model to sound more confident
D. Use it, because it looks precise

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q2
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LLM.04 (local)

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An answer explains a tax rule in formal language. Which step best checks it?

A. Read it again to see if it still sounds right
B. Trust it because tax rules are simple
C. Compare it with the official government page
D. Check whether it uses legal vocabulary

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q3
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Which answer feature is the best sign you can verify a claim?

A. It is longer than other answers
B. It uses many technical words
C. It is written in a calm, confident tone
D. It names a specific source you can look up

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q4
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.04 (local)

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A model confidently states a store's holiday hours. What is the most reliable next step?

A. Assume all stores have the same hours
B. Check the store's own website or call it
C. Ask the model to repeat it
D. Trust it because it sounded sure

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F1-Q5
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.04 (local)

Learn
Some answers include their own reassurances, such as "I am confident," "this is a verified fact," or a neat reference list. These are still generated text, so they do not add evidence by themselves. An answer can also mix correct and incorrect parts, which makes errors harder to notice: three right facts can make a fourth wrong one seem trustworthy. A practical habit is to identify the specific claims that matter for your decision and check those, rather than judging the whole answer by how it sounds.

An answer ends with "This is a verified fact." What does that sentence add?

A. No evidence; it is also generated text
B. Proof that a person verified it first
C. A legal guarantee of accuracy
D. A link to the verification

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q1
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.04 (local)

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An answer gives four facts; you confirm three are right. What should you assume about the fourth?

A. It must be wrong since the others were right
B. It can be ignored without checking
C. It must be right since the others were
D. Nothing yet; check it too if it matters

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q2
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.04 (local)

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You are deciding whether to buy a used car. A model summary of the car's history is long and fluent. Which claims should you check first?

A. None, since a long summary covers everything
B. The sentence with the friendliest tone
C. The claims the decision depends on, like accidents
D. Only the first sentence, which sets the tone for the rest

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q3
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LLM.04 (local)

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An answer includes a tidy reference list. What should you do before relying on it?

A. Open the references to see if they exist and agree
B. Count the references; more means more accurate
C. Trust it, since a reference list proves accuracy
D. Delete the list, since references are never real

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q4
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LLM.04 (local)

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Why can three correct facts make a fourth wrong fact more dangerous?

A. Readers always check the fourth fact first
B. The correct facts make the wrong one seem trustworthy
C. Three correct facts make all facts correct
D. Wrong facts disappear when mixed with right ones

REFERENCE LABEL: NOT PART OF THE QUESTION ID S17-F2-Q5
Artificial Intelligence, Set 17: Fluent is not the same as true · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.04 (local)

Set 18: Context windows

Outcome AI.LLM.05: Explain a context window as the limited amount of text a model can consider at once and predict what happens when a conversation or document exceeds it

Module: Foundation 3: Large language models

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Learn
A model can only consider a limited amount of text at once, called its CONTEXT WINDOW, measured in tokens. The context usually includes instructions the product adds, the conversation so far, any files or pasted text, and the reply being written. When a long conversation or document exceeds the window, something has to give: depending on the product, older messages may be dropped or shortened, part of a file may be left out, or the request may be refused. The model cannot use text that is no longer in its context, so it may "forget" early details or miss the end of a long document.

What is a context window?

A. The time of day when the model is available
B. The limited amount of text a model can use at once
C. The window on screen where the app is shown
D. The list of every fact the model learned

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q1
Artificial Intelligence, Set 18: Context windows · MAIN · Q1/5 · Outcome AI.LLM.05 (local)

Learn
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Which of these is usually part of the context?

A. The model's entire training data
B. Every conversation all users have ever had
C. The conversation so far and any pasted text
D. The user's whole computer hard drive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q2
Artificial Intelligence, Set 18: Context windows · MAIN · Q2/5 · Outcome AI.LLM.05 (local)

Learn
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A very long conversation exceeds the window. What might happen, depending on the product?

A. Older messages may be dropped or shortened
B. The model becomes permanently smarter
C. The model rewrites its training data
D. All messages are kept forever with no limit

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q3
Artificial Intelligence, Set 18: Context windows · MAIN · Q3/5 · Outcome AI.LLM.05 (local)

Learn
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Why might a model "forget" a detail from the start of a very long chat?

A. It chose to ignore the user on purpose
B. Models cannot read the first message ever
C. Details from chats are erased from the internet
D. That detail may no longer be in its context

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q4
Artificial Intelligence, Set 18: Context windows · MAIN · Q4/5 · Outcome AI.LLM.05 (local)

Learn
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In what unit is a context window usually measured?

A. Megapixels
B. Tokens
C. Minutes
D. Pages

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-M-Q5
Artificial Intelligence, Set 18: Context windows · MAIN · Q5/5 · Outcome AI.LLM.05 (local)

Learn
Imagine a WHITEBOARD in a meeting room. Everything the group is discussing must be written on it to be used. The board is big but not endless. When it fills up, someone erases the oldest notes to make room, and from then on the group cannot look at what was erased. A model's context window works like the board: instructions, the conversation and any pasted text are "written" there. When it fills, older material may be removed, and the model can no longer use it.

In the comparison, what is the whiteboard like?

A. The model's training data
B. The user's keyboard
C. The company office
D. The context window

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q1
Artificial Intelligence, Set 18: Context windows · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LLM.05 (local)

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What happens when the board fills up?

A. All notes are kept
B. The board grows forever
C. The oldest notes are erased
D. The meeting ends at once

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q2
Artificial Intelligence, Set 18: Context windows · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LLM.05 (local)

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Can the group use notes that were erased?

A. No, they are gone from the board
B. Yes, if they shout
C. Only on Fridays
D. Yes, the board remembers them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q3
Artificial Intelligence, Set 18: Context windows · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LLM.05 (local)

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What is written on the board?

A. Only the newest single word
B. Instructions, the chat and pasted text
C. Nothing at all
D. Everything ever said anywhere in the building

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q4
Artificial Intelligence, Set 18: Context windows · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LLM.05 (local)

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Why might the group forget a point made at the start?

A. It may have been erased to make room
B. Points made first are always wrong
C. It was never said
D. The board was too small to write on at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-B1-Q5
Artificial Intelligence, Set 18: Context windows · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LLM.05 (local)

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After two hours of chat, a model ignores a formatting rule Kim gave at the start. What should she try?

A. Assume the model is broken forever

B. Delete her account

C. Restate the rule in a new message

D. Wait until tomorrow

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q1
Artificial Intelligence, Set 18: Context windows · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LLM.05 (local)

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Omar pastes a 400-page manual, and the answer seems to miss the last chapters. What is a likely reason?

A. The last chapters were written in code
B. The model already knew those chapters
C. Models refuse to read past page 100 by law
D. Part of the manual may not fit in the context

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q2
Artificial Intelligence, Set 18: Context windows · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LLM.05 (local)

Learn
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Which approach helps with a very long document?

A. Paste the whole manual three times to be safe
B. Split it into sections and do one at a time
C. Remove all headings first
D. Send it with no question at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q3
Artificial Intelligence, Set 18: Context windows · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LLM.05 (local)

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A product says its context window is 8,000 tokens. Using about three quarters of a word per token, about how many words of total conversation can it consider?

A. About 6,000 words
B. About 2,000 words
C. About 10,667 words
D. Exactly 8,000 words

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q4
Artificial Intelligence, Set 18: Context windows · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.05 (local)

Learn
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Why does starting a fresh conversation sometimes fix confused answers?

A. It deletes the model's parameters
B. It switches the user to a paid account
C. It retrains the model from scratch
D. It clears old text that crowded the context

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F1-Q5
Artificial Intelligence, Set 18: Context windows · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.05 (local)

Learn
Even text that fits in the context window is not always used equally well. Models can overlook details buried in the middle of a very long input, especially when many similar details compete. Practical habits help: put the most important instructions and facts where they are easy to find, restate key requirements when a conversation gets long, split very long documents into sections, and check that answers cover the parts you care about. A bigger context window raises the limit; it does not guarantee careful reading.

A long report fits in the context, but the answer misses a key figure in the middle. What does this show?

A. Models never read the middle of documents
B. The figure must not exist in the report
C. Fitting in the window does not ensure careful use
D. Context windows are always unlimited

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q1
Artificial Intelligence, Set 18: Context windows · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.05 (local)

Learn
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Where should you put the most important instructions in a long request?

A. Only in the file name
B. Somewhere easy to find, then restate them
C. Hidden in the middle among many other details
D. Nowhere; models guess instructions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q2
Artificial Intelligence, Set 18: Context windows · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.05 (local)

Learn
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A tool advertises a much bigger context window. What can you conclude?

A. It must use a completely different kind of model
B. It now reads every detail perfectly
C. It no longer makes any errors
D. It takes more text; answers still need checking

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q3
Artificial Intelligence, Set 18: Context windows · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LLM.05 (local)

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You need answers about sections 2 and 7 of a long contract. What is a sensible way to check the answer?

A. Confirm it covers both sections, using the text
B. Check only section 2
C. Accept it as long as the answer is long and detailed
D. Assume both were read

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q4
Artificial Intelligence, Set 18: Context windows · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LLM.05 (local)

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A conversation has many similar product codes. The model mixes two up. What would help?

A. Change the font of the codes
B. Add even more similar codes
C. Restate the key codes clearly in this message
D. Ask the model to sound more confident about them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S18-F2-Q5
Artificial Intelligence, Set 18: Context windows · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.05 (local)

Set 19: What a model learned versus what you supply

Outcome AI.LLM.06: Tell knowledge learned in training (with a cutoff date) from information supplied or retrieved during a request

Module: Foundation 3: Large language models

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A model's built-in knowledge comes from its training data, which stops at a TRAINING CUTOFF date. It does not know about events after that date unless the information is SUPPLIED or RETRIEVED during your request: you paste a document, or the product runs a web search and adds the results to the context. These are different: built-in knowledge is patterns frozen at training time and may be incomplete or wrong, while supplied or retrieved text is current material the model reads now, which is only as good as its source. When an answer depends on recent facts, know which kind you are getting.

What is a training cutoff?

A. The date after which the training data has no information
B. The time limit for typing a question
C. The point where a model stops answering for the day
D. The maximum number of users at once

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q1
Artificial Intelligence, Set 19: What a model learned versus what you supply · MAIN · Q1/5 · Outcome AI.LLM.06 (local)

Learn
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A model's training cutoff was before a recent election. How could it still describe the result correctly?

A. By training itself during the conversation
B. If the result is supplied or retrieved for it
C. It cannot ever do so in any situation
D. By guessing based on old patterns

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q2
Artificial Intelligence, Set 19: What a model learned versus what you supply · MAIN · Q2/5 · Outcome AI.LLM.06 (local)

Learn
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A product runs a web search and adds the results to the request. Which kind of information is that?

A. Information that has no source at all
B. Built-in knowledge from training
C. A permanent change to the model
D. Retrieved information read now

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q3
Artificial Intelligence, Set 19: What a model learned versus what you supply · MAIN · Q3/5 · Outcome AI.LLM.06 (local)

Learn
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Which statement about retrieved information is accurate?

A. It is the same as training data
B. It permanently updates the model's knowledge
C. It is only as good as the source it came from
D. It is always correct because it is recent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q4
Artificial Intelligence, Set 19: What a model learned versus what you supply · MAIN · Q4/5 · Outcome AI.LLM.06 (local)

Learn
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Which question most needs supplied or retrieved information rather than built-in knowledge?

A. What is a triangle?
B. What is the boiling point of water at sea level?
C. What is a noun in English grammar?
D. What are this week's opening hours for the city pool?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-M-Q5
Artificial Intelligence, Set 19: What a model learned versus what you supply · MAIN · Q5/5 · Outcome AI.LLM.06 (local)

Learn
A printed ENCYCLOPEDIA on a shelf holds a lot of knowledge, but only up to the year it was printed. It cannot tell you yesterday's news. If you also hand someone TODAY'S NEWSPAPER, they can read current events, but the newspaper might contain mistakes, and it does not change the encyclopedia. A model's built-in knowledge is like the encyclopedia: fixed at its training cutoff. Supplied or retrieved text is like the newspaper: current material read right now.

The encyclopedia is like which kind of information?

A. Retrieved information
B. Built-in knowledge
C. A web search
D. Neither kind

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q1
Artificial Intelligence, Set 19: What a model learned versus what you supply · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LLM.06 (local)

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Today's newspaper is like which kind of information?

A. Supplied or retrieved text
B. The model's trained parameters
C. Training data
D. Built-in knowledge

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q2
Artificial Intelligence, Set 19: What a model learned versus what you supply · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LLM.06 (local)

Learn
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Can the encyclopedia tell you yesterday's news?

A. Only on weekends
B. Yes, if you read it slowly
C. No, it stops at its printing date
D. Yes, encyclopedias update themselves

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q3
Artificial Intelligence, Set 19: What a model learned versus what you supply · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LLM.06 (local)

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Does reading the newspaper change the encyclopedia?

A. Yes, it rewrites the pages
B. No, it stays the same
C. Yes, it adds a new volume
D. Only if the news is big

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q4
Artificial Intelligence, Set 19: What a model learned versus what you supply · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LLM.06 (local)

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Is the newspaper guaranteed to be correct?

A. Yes, because it is new
B. Yes, all news is correct
C. Only if it is long
D. No, it can contain mistakes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-B1-Q5
Artificial Intelligence, Set 19: What a model learned versus what you supply · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LLM.06 (local)

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Ben asks which phone model is newest. The tool does not search the web. What should he expect?

A. It will always name the newest model released today
B. It will refuse to mention any phones
C. The answer may be out of date, reflecting its training cutoff
D. It will search stores secretly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q1
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LLM.06 (local)

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A student pastes this year's club rules and asks about meeting times. What is the model using?

A. The supplied rules in the current request
B. A web search of the school's site
C. Its memory of every club in the world
D. A retrained version of itself

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q2
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LLM.06 (local)

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Which situation shows retrieval?

A. The user types a short question with no files
B. The product searches news sites and quotes them
C. The model answers from what it learned in training
D. The model writes a poem from imagination

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q3
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LLM.06 (local)

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A retrieved web page is three years old. What should you do before relying on it?

A. Ask the model to make it newer
B. Trust it, because it was retrieved by the product
C. Assume nothing has changed
D. Check a current source for time-sensitive facts

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q4
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.06 (local)

Learn
A model's built-in knowledge comes from its training data, which stops at a TRAINING CUTOFF date. It does not know about events after that date unless the information is SUPPLIED or RETRIEVED during your request: you paste a document, or the product runs a web search and adds the results to the context. These are different: built-in knowledge is patterns frozen at training time and may be incomplete or wrong, while supplied or retrieved text is current material the model reads now, which is only as good as its source. When an answer depends on recent facts, know which kind you are getting.

Which question can a model usually answer from built-in knowledge without current sources?

A. Is the corner store open this holiday weekend?
B. Who won last night's game?
C. What does the word "photosynthesis" mean?
D. What is today's exchange rate?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F1-Q5
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.06 (local)

Learn
Answers often blend both kinds of information, and it is not always obvious which part came from where. A model might quote a supplied article correctly but fill a gap with older built-in knowledge, or describe a "latest" product that has since been replaced. Retrieved sources can also be outdated, unreliable or off-topic. Useful habits: ask the answer to say which statements come from the supplied material, check the dates of any sources, and confirm time-sensitive facts such as prices, laws, schedules and officials directly.

An answer quotes a supplied article correctly but adds a detail the article does not contain. Where might that detail have come from?

A. Older built-in knowledge or an invented guess
B. The supplied article, which must contain it
C. A law that requires extra details
D. A permanent update to the model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q1
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.06 (local)

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You want to know which parts of an answer came from your uploaded document. What is a useful request?

A. Ask it to sound more confident about everything
B. Ask it to stop using the document
C. Ask it to mark statements taken from the document
D. Ask for a longer answer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q2
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.06 (local)

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A model describes "the latest version" of a product. How should you treat that phrase?

A. Check, since "latest" may reflect an old cutoff
B. Trust it, since models always know the latest
C. Assume no newer version can exist
D. Assume it is always wrong

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q3
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LLM.06 (local)

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A retrieved source is an unknown blog with no author or date. What is the best step?

A. Trust it, since the product chose to retrieve it
B. Use it but remove the link
C. Assume blogs are always wrong
D. Confirm it with a more reliable, dated source

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q4
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LLM.06 (local)

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Which facts most need direct confirmation even after retrieval?

A. The meaning of common words
B. Prices, laws, schedules and officeholders
C. The plot of a famous novel from long ago
D. Basic arithmetic facts

REFERENCE LABEL: NOT PART OF THE QUESTION ID S19-F2-Q5
Artificial Intelligence, Set 19: What a model learned versus what you supply · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.06 (local)

Set 20: Conversation context versus saved memory

Outcome AI.LLM.07: Tell the text in the current conversation from product features that save information between conversations, and check a product's own settings

Module: Foundation 3: Large language models

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Two different things can make a chatbot seem to "remember." CONVERSATION CONTEXT is the text of the current chat that is sent back to the model with each message, so it can refer to what you said earlier in that chat. SAVED MEMORY (or history features) is something a PRODUCT may offer: it stores notes or past chats and adds them to future conversations. Saved memory is a product feature, it varies between products, and it can usually be viewed, turned off or deleted in settings. Neither one retrains the model. Check what a specific product saves rather than assuming.

In one chat, you mention your dog's name, and later in the same chat the model uses it. What makes this possible?

A. The earlier message is still in the chat context
B. The model was retrained on your dog's name
C. The model guessed the name from training data
D. Every model knows every pet's name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q1
Artificial Intelligence, Set 20: Conversation context versus saved memory · MAIN · Q1/5 · Outcome AI.LLM.07 (local)

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A product greets you by name in a brand-new chat. What is the most likely reason?

A. The model's parameters changed to include your name
B. The model read your mind from your typing
C. All chatbots know every user's name
D. The product saved a note and added it to the new chat

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q2
Artificial Intelligence, Set 20: Conversation context versus saved memory · MAIN · Q2/5 · Outcome AI.LLM.07 (local)

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Which statement about saved memory is accurate?

A. It is part of every language model's design
B. It can never be viewed or deleted
C. It is a product feature that varies and has settings
D. It permanently retrains the model for every user

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q3
Artificial Intelligence, Set 20: Conversation context versus saved memory · MAIN · Q3/5 · Outcome AI.LLM.07 (local)

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Does conversation context retrain the model?

A. Yes; every chat retrains the model
B. No; it is just text sent with each message
C. Yes; but only on weekends
D. No; because context is deleted instantly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q4
Artificial Intelligence, Set 20: Conversation context versus saved memory · MAIN · Q4/5 · Outcome AI.LLM.07 (local)

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A friend says, "All chatbots remember everything forever." What is the best response?

A. True; models store chats in their parameters
B. False; no chatbot can ever save anything
C. It depends on the product; check its memory settings
D. True; every chatbot saves all chats forever

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-M-Q5
Artificial Intelligence, Set 20: Conversation context versus saved memory · MAIN · Q5/5 · Outcome AI.LLM.07 (local)

Learn
Talking to a friend at lunch, you can say "the movie I mentioned" because it came up a few minutes ago: that is like CONVERSATION CONTEXT. If your friend also keeps a NOTEBOOK and writes "likes horror movies," then next month they can bring it up again: that is like SAVED MEMORY. The notebook is something extra that your friend chose to keep, could show you, and could erase. Neither the lunch talk nor the notebook changes who your friend fundamentally is, just as neither one retrains a model.

Saying "the movie I mentioned" during lunch is like what?

A. A web search
B. Conversation context
C. Saved memory
D. Training a new model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q1
Artificial Intelligence, Set 20: Conversation context versus saved memory · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LLM.07 (local)

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The friend's notebook is like what?

A. Saved memory
B. Conversation context
C. The model's parameters
D. A search engine

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q2
Artificial Intelligence, Set 20: Conversation context versus saved memory · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LLM.07 (local)

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Can the friend show you or erase the notebook?

A. Only if it is full
B. No, it is secret forever
C. No, notebooks like that can never be erased or shown
D. Yes, it is kept separately and can be managed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q3
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Next month, how could the friend recall "likes horror movies"?

A. By guessing
B. By asking a stranger
C. By reading the notebook
D. By remembering every word

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Does keeping a notebook change who the friend is?

A. No, it is extra information they keep
B. Only if the notebook is long
C. Yes, they become a new person entirely
D. Yes, it rewires their brain

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-B1-Q5
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Lia wants a product to stop bringing up her old projects in new chats. What should she check?

A. Whether the model needs retraining
B. The product's memory or history settings
C. Whether her internet is slow
D. Whether her messages are too short

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In a long chat, a model correctly refers to a recipe from 20 minutes earlier. What does it rely on?

A. Every recipe ever written
B. A permanent change to its parameters
C. A separate saved note from last year
D. The conversation context from this chat

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q2
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A product has no memory feature. You start a new chat. Should you expect it to know your earlier chat?

A. No; and it will refuse to talk to you
B. No; you would need to supply the details again
C. Yes; every model remembers every past chat
D. Yes; the earlier chat retrained it

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Which action is about saved memory rather than conversation context?

A. Viewing and deleting stored notes in settings
B. Scrolling up to reread this chat
C. Pasting a file into this message
D. Restating a rule in the current chat window

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q4
Artificial Intelligence, Set 20: Conversation context versus saved memory · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.07 (local)

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A coworker says, "I told it my preference once, so the model learned it forever." What is more accurate?

A. Correct; all preferences are added to its training right away
B. Wrong; products never save preferences
C. A product may have saved a note; the model did not retrain
D. Correct; models retrain after each preference

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F1-Q5
Artificial Intelligence, Set 20: Conversation context versus saved memory · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.07 (local)

Learn
Memory features raise practical questions. Something you mentioned weeks ago could appear in a new answer, which may be helpful or unwelcome. Shared devices or shared accounts can expose saved notes to other people. Turning off memory may not delete chats already stored, and deleting a chat may not remove information a product saved separately; read what each setting actually does. For sensitive topics, the safest habit is not to share details you would not want stored, whatever the settings say.

A family shares one account, and a new answer mentions a private detail one member shared. What is the issue?

A. The model hacked the family's computer
B. Memory features never store private details
C. This is impossible with any product
D. Saved memory on a shared account can expose details

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q1
Artificial Intelligence, Set 20: Conversation context versus saved memory · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.07 (local)

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Jay turns memory off. Does that automatically delete chats already stored?

A. No, chats can never be deleted by anyone
B. Not necessarily; read what the setting does
C. Yes, always, in every product
D. Yes, and it retrains the model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q2
Artificial Intelligence, Set 20: Conversation context versus saved memory · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.07 (local)

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Which is the safest habit for very sensitive details, such as medical history?

A. Share them with a friendly greeting
B. Share them, since settings always protect you
C. Share them only in long conversations
D. Do not share details you would not want stored

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q3
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A user deletes a chat but later sees a detail from it in a new answer. What is a possible explanation?

A. The new chat is the same chat
B. The detail was in the model's original training
C. The product saved that detail separately as a note
D. Deleting chats retrains the model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q4
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What is the best source for how a specific product handles memory?

A. Its own settings and help pages
B. A general article about all chatbots
C. A friend's guess
D. The model's answer about itself

REFERENCE LABEL: NOT PART OF THE QUESTION ID S20-F2-Q5
Artificial Intelligence, Set 20: Conversation context versus saved memory · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.07 (local)

Set 21: A product is more than its model

Outcome AI.LLM.09: Explain that a product adds its own instructions, settings, tools and limits, so a general description of LLMs cannot predict one product's behavior

Module: Foundation 3: Large language models

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Learn
A general description of language models cannot tell you exactly how a particular product will behave. Products add their own layers: SYSTEM INSTRUCTIONS the company writes (for example, "be brief" or "do not give medical diagnoses"), SAFETY FILTERS, TOOLS such as web search, code running or file reading, SETTINGS, and LIMITS on length or usage. Two products built on similar models can therefore refuse different requests, format answers differently, or have different abilities. To know what a product does, test it on your task and read its documentation.

What are system instructions?

A. Questions the user types into the chat box
B. Directions the company adds to shape answers
C. The operating system of your phone
D. The training data used to build the model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q1
Artificial Intelligence, Set 21: A product is more than its model · MAIN · Q1/5 · Outcome AI.LLM.09 (local)

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Two products use similar models, but one refuses a request the other accepts. What is a likely reason?

A. Different instructions or filters in each product
B. Similar models always behave identically in every product
C. The user typed the request too slowly
D. One model must be completely broken

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q2
Artificial Intelligence, Set 21: A product is more than its model · MAIN · Q2/5 · Outcome AI.LLM.09 (local)

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A product can read your spreadsheet files. Where does that ability come from?

A. The model memorized your spreadsheet in training
B. Every language model reads files by itself
C. Your spreadsheet program sends answers automatically
D. A file-reading tool the product provides

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q3
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What is the best way to learn how a specific product behaves?

A. Read a general description of all language models
B. Ask a friend who uses a different product
C. Test it on your task and read its documentation
D. Assume it matches the first chatbot you used

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q4
Artificial Intelligence, Set 21: A product is more than its model · MAIN · Q4/5 · Outcome AI.LLM.09 (local)

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Which is an example of a product limit?

A. A maximum length for each answer
B. The meaning of a word
C. The user's choice of topic
D. The patterns learned in training

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-M-Q5
Artificial Intelligence, Set 21: A product is more than its model · MAIN · Q5/5 · Outcome AI.LLM.09 (local)

Learn
Two restaurants might buy ingredients from the same supplier, yet their meals differ: each has its own RECIPES, its own HOUSE RULES ("no substitutions"), its own EQUIPMENT (one has a pizza oven), and its own PORTION SIZES. Knowing the supplier tells you little about what you will be served. An AI product is similar: the model is like the supplier's ingredients, while the company's instructions, filters, tools and limits are like the recipes, house rules, equipment and portions.

In the comparison, the model is like what?

A. The pizza oven
B. The house rules
C. The supplier's ingredients
D. The portion sizes it serves

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q1
Artificial Intelligence, Set 21: A product is more than its model · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.LLM.09 (local)

Learn
Two restaurants might buy ingredients from the same supplier, yet their meals differ: each has its own RECIPES, its own HOUSE RULES ("no substitutions"), its own EQUIPMENT (one has a pizza oven), and its own PORTION SIZES. Knowing the supplier tells you little about what you will be served. An AI product is similar: the model is like the supplier's ingredients, while the company's instructions, filters, tools and limits are like the recipes, house rules, equipment and portions.

A "no substitutions" rule is like which product layer?

A. The model's original training
B. System instructions or filters
C. The supplier
D. The user's order

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q2
Artificial Intelligence, Set 21: A product is more than its model · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.LLM.09 (local)

Learn
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One restaurant has a pizza oven. In AI terms, that is like what?

A. A safety filter
B. The model's learned parameters
C. Training data
D. A tool the product provides

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q3
Artificial Intelligence, Set 21: A product is more than its model · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.LLM.09 (local)

Learn
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Why can two restaurants using the same supplier serve different meals?

A. The supplier changes ingredients for each one
B. Their recipes, rules, equipment and portions differ
C. Customers cook the meals themselves
D. They cannot; same supplier means same meals

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q4
Artificial Intelligence, Set 21: A product is more than its model · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.LLM.09 (local)

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Knowing the supplier tells you how much about the meal?

A. Everything about it
B. Nothing at all
C. Exactly the price
D. Only a little

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-B1-Q5
Artificial Intelligence, Set 21: A product is more than its model · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.LLM.09 (local)

Learn
A general description of language models cannot tell you exactly how a particular product will behave. Products add their own layers: SYSTEM INSTRUCTIONS the company writes (for example, "be brief" or "do not give medical diagnoses"), SAFETY FILTERS, TOOLS such as web search, code running or file reading, SETTINGS, and LIMITS on length or usage. Two products built on similar models can therefore refuse different requests, format answers differently, or have different abilities. To know what a product does, test it on your task and read its documentation.

A school's chatbot refuses to write complete essays, but a public chatbot will. What explains this best?

A. The school product added limits on essay writing
B. Essays cannot be generated by any model
C. The school chatbot cannot write sentences at all
D. Public chatbots ignore all instructions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q1
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.LLM.09 (local)

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An article says "language models cannot browse the web." Your product shows web results. What is true?

A. Browsing is part of every model
B. The article proves your product is fake
C. Your product adds a browsing tool to its model
D. The model learned the web results in training

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q2
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.LLM.09 (local)

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Before relying on a product to summarize contracts, what should a manager do?

A. Trust a general article about language models
B. Test it on sample contracts and read its documentation
C. Assume all products summarize equally well
D. Ask the product if it is good at contracts

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q3
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.LLM.09 (local)

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A product gives very short answers no matter what you ask. What is a likely reason?

A. All language models only give short answers
B. Your device has a broken keyboard
C. Its instructions or settings limit answer length
D. The model has run out of words forever

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q4
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.LLM.09 (local)

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Which statement is most accurate?

A. Products built on similar models can behave quite differently
B. Products never change how models behave
C. A model description predicts product behavior exactly
D. Any two products with similar models act the same

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F1-Q5
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.LLM.09 (local)

Learn
Products also change. A company may update instructions, add or remove tools, adjust filters, or switch models, sometimes without obvious notice. A workflow that worked last month can break, and a limit that blocked you may disappear. Version names, release notes and help pages are the best record. When results matter, keep a few test cases and rerun them after updates, and avoid building important processes on an undocumented behavior you happened to notice once.

A workflow that summarized meeting notes worked last month but now formats answers differently. What is a likely explanation?

A. Your computer's clock is wrong
B. The workflow was never possible
C. The meeting notes became illegal to summarize
D. The product updated its instructions or model

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q1
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.LLM.09 (local)

Learn
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Where is the best record of what changed in a product?

A. Its release notes and help pages
B. A guess from a friend
C. An old screenshot from last year
D. The model's description of itself

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q2
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.LLM.09 (local)

Learn
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A team relies on a product for weekly reports. What habit protects them from surprises?

A. Switch products every week
B. Never look at the outputs again once it works
C. Assume updates never affect results
D. Keep test cases and rerun them after updates

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q3
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.LLM.09 (local)

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You noticed once that a product ignores a filter if you phrase things oddly. Should you build a process on this?

A. No; because products never change
B. No; undocumented behavior can stop at any time
C. Yes; odd phrasing is an official feature of the product
D. Yes; once noticed, it will always work

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q4
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.LLM.09 (local)

Learn
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A limit that blocked a request last month is gone. What is the most accurate conclusion?

A. The request must now be unsafe
B. Limits never existed
C. The product changed its limits; check its documentation
D. The model learned on its own to ignore its limits

REFERENCE LABEL: NOT PART OF THE QUESTION ID S21-F2-Q5
Artificial Intelligence, Set 21: A product is more than its model · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.LLM.09 (local)

Set 22: Task, context, audience and purpose

Outcome AI.PROMPT.01: Identify the task, context, audience and purpose a request needs, and spot which one is missing

Module: Foundation 4: Asking for what you need

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A useful request (often called a PROMPT) usually covers four things. The TASK: what you want done ("write," "compare," "list," "explain"). The CONTEXT: background the tool cannot know, such as your situation, the facts involved or what you already tried. The AUDIENCE: who will read or use the result, such as a ten-year-old, a manager or yourself. The PURPOSE: why you need it, such as to persuade, to decide or to study. A vague request like "write about recycling" leaves the tool to guess all four, so the result may be generic. Adding the missing pieces gives it a fair chance.

In "Explain (the action word) photosynthesis (the process) to a group of eight-year-olds," which part names the audience?

A. to
B. photosynthesis (the process)
C. Explain (the action word)
D. a group of eight-year-olds

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q1
Artificial Intelligence, Set 22: Task, context, audience and purpose · MAIN · Q1/5 · Outcome AI.PROMPT.01 (local)

Learn
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"Write about recycling." Which pieces are missing?

A. Only the task, which is not stated at all
B. Context, audience and purpose
C. Nothing is missing; the request is complete
D. Only the topic of the writing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q2
Artificial Intelligence, Set 22: Task, context, audience and purpose · MAIN · Q2/5 · Outcome AI.PROMPT.01 (local)

Learn
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Which detail is CONTEXT the tool could not know on its own?

A. "Our school recycles paper, not plastic."
B. "Please answer in English."
C. "Make it sound friendly and clear."
D. "Use short sentences in the reply."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q3
Artificial Intelligence, Set 22: Task, context, audience and purpose · MAIN · Q3/5 · Outcome AI.PROMPT.01 (local)

Learn
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Which phrase states a PURPOSE?

A. "about 200 words long"
B. "for the class newsletter next week"
C. "so the class votes for a compost bin"
D. "write a short article"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q4
Artificial Intelligence, Set 22: Task, context, audience and purpose · MAIN · Q4/5 · Outcome AI.PROMPT.01 (local)

Learn
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Why does a vague request often get a generic answer?

A. Short requests use too many tokens
B. Vague requests are always rejected by filters
C. Tools refuse to answer short requests
D. The tool has to guess the missing pieces

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-M-Q5
Artificial Intelligence, Set 22: Task, context, audience and purpose · MAIN · Q5/5 · Outcome AI.PROMPT.01 (local)

Learn

Imagine asking a new babysitter to "make dinner." That is a REQUEST, but the babysitter does not know the BACKGROUND: one child is allergic to nuts, dinner is at 6, and the kids love pasta. Without that background, the babysitter might make something unsafe or unwanted. An AI tool is similar. The request says what to do; the background, or context, says what the tool cannot see for itself. Good results need both, but a request alone often feels complete to the person asking, because they already know the background.

In "make dinner," what is missing?

A. The background, such as allergies

B. The house address

C. The request itself

D. The babysitter's full name and age

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q1

Artificial Intelligence, Set 22: Task, context, audience and purpose · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.PROMPT.01 (local)

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"One child is allergic to nuts" is which part?

A. A greeting
B. The request
C. Background (context)
D. The audience for dinner

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q2
Artificial Intelligence, Set 22: Task, context, audience and purpose · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.PROMPT.01 (local)

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Why might the request feel complete to the parent?

A. Babysitters can read minds
B. The parent already knows the background
C. Requests never need details
D. The kids will explain everything to the babysitter

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q3
Artificial Intelligence, Set 22: Task, context, audience and purpose · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.PROMPT.01 (local)

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What could happen without the background?

A. An unsafe or unwanted dinner
B. The kids cooking instead
C. A perfect dinner every time
D. No dinner at all, ever

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q4
Artificial Intelligence, Set 22: Task, context, audience and purpose · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.PROMPT.01 (local)

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In an AI request, what does context provide?

A. The tool's price
B. Facts the tool cannot see for itself
C. The tool's training data
D. A list of the tool's display settings

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-B1-Q5
Artificial Intelligence, Set 22: Task, context, audience and purpose · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.PROMPT.01 (local)

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Pat needs a short email asking her landlord to fix a leaking sink before Friday. Which request is best?

A. "Explain how sinks leak and how plumbers fix them."
B. "Write something about landlords in general."
C. "Draft a polite email asking my landlord to fix the sink by Friday."
D. "Write an email to my landlord about things in my apartment."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q1
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.PROMPT.01 (local)

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A student wants a study guide for a biology test on cells. What is the most useful detail to add to "Make a study guide"?

A. The color of the student's notebook
B. A request to make the guide sound impressive
C. The name of the student's best friend
D. The topic, the course level and the test format

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q2
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.PROMPT.01 (local)

Learn
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Which request names a clear audience?

A. "Explain credit scores in some detail for me."
B. "Explain credit scores very well, please."
C. "Explain credit scores for a high-school senior."
D. "Explain credit scores, and make it good."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q3
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.PROMPT.01 (local)

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Marco asks, "Give me ideas." What should he add first?

A. A request for the ideas to be numbered in bold
B. Nothing; short requests always work best
C. The word "please" at the start of the sentence
D. What the ideas are for and any limits

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q4
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.PROMPT.01 (local)

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Which request includes context the tool could not guess?

A. "Plan meals for a week; we have no oven, only a stove."
B. "Plan a week of meals as fast as you can."
C. "Plan a week of meals, please and thank you."
D. "Plan meals for a week that taste really good."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F1-Q5
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.PROMPT.01 (local)

Learn
Sometimes a request seems complete but the pieces pull against each other, or one key detail is assumed rather than stated. "Explain this contract simply for my grandmother, in full legal detail" asks for two different audiences at once. "Fix the schedule" assumes the tool knows which schedule and what is wrong with it. Before sending an important request, reread it as a stranger would: could someone with no background do exactly what you want? If not, add or fix the missing piece first.

"Explain this contract simply for my grandmother, in full legal detail." What is the problem?

A. Grandmothers cannot read contracts
B. It asks for two different audiences at once
C. It is too short to be understood at all
D. It does not say what the contract is about

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q1
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.PROMPT.01 (local)

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"Fix the schedule." What does this request assume?

A. That the tool knows which schedule and what is wrong
B. That the tool can see every calendar in the world
C. That schedules never need fixing at all
D. That the user wants a poem

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q2
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.PROMPT.01 (local)

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What is a good test before sending an important request?

A. Count the words and make sure it is very short
B. Reread it as a stranger with no background would
C. Send it and hope for the best
D. Add the word "urgent" to the start

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q3
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.PROMPT.01 (local)

Learn
Sometimes a request seems complete but the pieces pull against each other, or one key detail is assumed rather than stated. "Explain this contract simply for my grandmother, in full legal detail" asks for two different audiences at once. "Fix the schedule" assumes the tool knows which schedule and what is wrong with it. Before sending an important request, reread it as a stranger would: could someone with no background do exactly what you want? If not, add or fix the missing piece first.

"Summarize the report for the board so they can decide on the budget." Which piece is still missing?

A. The purpose, which is not mentioned anywhere
B. The task, which is not mentioned anywhere
C. Which report (it must be supplied)
D. The audience, which is not mentioned anywhere

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q4
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.PROMPT.01 (local)

Learn
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A request asks for "a fun, serious speech." What should the writer do first?

A. Send it as written; tools resolve conflicts well
B. Ask for ten speeches at once
C. Remove the word "speech" from the request
D. Decide which tone matters more, or how to balance them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S22-F2-Q5
Artificial Intelligence, Set 22: Task, context, audience and purpose · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.PROMPT.01 (local)

Set 23: Output format and constraints

Outcome AI.PROMPT.02: State the output format and constraints (length, tone, things to avoid) that a result must meet

Module: Foundation 4: Asking for what you need

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Besides saying what you want, say what the result should LOOK LIKE. The OUTPUT FORMAT is its shape: a bulleted list, a table with named columns, a three-paragraph letter, a numbered set of steps. CONSTRAINTS are limits it must respect: a length ("under 150 words"), a tone ("friendly but professional"), a reading level ("for beginners"), or things to avoid ("no jargon," "do not mention prices"). Stating format and constraints up front saves rounds of revision and makes the result easy to check: you can see at once whether it has five bullets or is under 150 words.

Which part of a request describes the OUTPUT FORMAT?

A. "compare three caterers"
B. "as a table with Item and Price columns"
C. "for my coworkers"
D. "so we can choose a caterer by Friday"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q1
Artificial Intelligence, Set 23: Output format and constraints · MAIN · Q1/5 · Outcome AI.PROMPT.02 (local)

Learn
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Which is a CONSTRAINT?

A. "write a cover letter"
B. "for a hiring manager"
C. "to apply for a job"
D. "under 150 words"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q2
Artificial Intelligence, Set 23: Output format and constraints · MAIN · Q2/5 · Outcome AI.PROMPT.02 (local)

Learn

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Fictional example (prepared in advance; not a real AI output)

Request:

> List three tips for saving money on groceries. Use a numbered list. Keep each tip under 15 words.

Response:

> 1. Plan your meals for the week before you go shopping.
> 2. Compare unit prices instead of package prices on the shelf.
> 3. Buy store brands, which are often cheaper than name brands, especially for basic items like flour, sugar, rice and canned vegetables.

In this fictional example, which instruction did the response NOT follow?

A. Giving three tips

B. Using a numbered list

C. Keeping each tip under 15 words

D. Keeping the topic on groceries

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q3

Artificial Intelligence, Set 23: Output format and constraints · MAIN · Q3/5 · Outcome AI.PROMPT.02 (local)

Learn
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Why does stating format up front help?

A. It saves revisions and makes checking easy
B. It makes the tool respond more slowly
C. It guarantees the facts are correct
D. It lets you skip reading the result

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q4
Artificial Intelligence, Set 23: Output format and constraints · MAIN · Q4/5 · Outcome AI.PROMPT.02 (local)

Learn
Besides saying what you want, say what the result should LOOK LIKE. The OUTPUT FORMAT is its shape: a bulleted list, a table with named columns, a three-paragraph letter, a numbered set of steps. CONSTRAINTS are limits it must respect: a length ("under 150 words"), a tone ("friendly but professional"), a reading level ("for beginners"), or things to avoid ("no jargon," "do not mention prices"). Stating format and constraints up front saves rounds of revision and makes the result easy to check: you can see at once whether it has five bullets or is under 150 words.

Which instruction is something to AVOID, a kind of constraint?

A. "Explain the field trip."
B. "Do not mention prices."
C. "Write it for parents."
D. "Make a checklist."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-M-Q5
Artificial Intelligence, Set 23: Output format and constraints · MAIN · Q5/5 · Outcome AI.PROMPT.02 (local)

Learn
A CHECKABLE instruction is one you can test by looking at the result. "Short" is vague: short to whom? "Under 100 words" is checkable: count and see. "Nice looking" is vague; "a table with three columns" is checkable. "Friendly" is partly vague, but "no technical terms" is easier to test. To turn a wish into a checkable instruction, ask: "What would I look for to know it was done?" Then write that down as the instruction.

Which instruction is checkable?

A. "Not too long"
B. "Brief-ish"
C. "Short"
D. "Under 100 words"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q1
Artificial Intelligence, Set 23: Output format and constraints · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.PROMPT.02 (local)

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Which is the most checkable version of "nice looking"?

A. "A table with three columns"
B. "Stylish"
C. "Pretty and neat"
D. "Very professional-looking"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q2
Artificial Intelligence, Set 23: Output format and constraints · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.PROMPT.02 (local)

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What question helps turn a wish into a checkable instruction?

A. "How do I sound polite?"
B. "Which tool is cheapest this month?"
C. "How would I know it was done?"
D. "How fast can it answer?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q3
Artificial Intelligence, Set 23: Output format and constraints · BACKWARD 1 (underlying skill) · Q3/5 · Outcome AI.PROMPT.02 (local)

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Which instruction is easier to test than "friendly"?

A. "No technical terms"
B. "Happy-sounding"
C. "Warm"
D. "Kind and fun"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q4
Artificial Intelligence, Set 23: Output format and constraints · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.PROMPT.02 (local)

Learn
A CHECKABLE instruction is one you can test by looking at the result. "Short" is vague: short to whom? "Under 100 words" is checkable: count and see. "Nice looking" is vague; "a table with three columns" is checkable. "Friendly" is partly vague, but "no technical terms" is easier to test. To turn a wish into a checkable instruction, ask: "What would I look for to know it was done?" Then write that down as the instruction.

Which result meets "exactly five bullet points"?

A. A paragraph with no bullets
B. A list with four bullets
C. A list with five bullets
D. A list with seven bullets

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-B1-Q5
Artificial Intelligence, Set 23: Output format and constraints · BACKWARD 1 (underlying skill) · Q5/5 · Outcome AI.PROMPT.02 (local)

Learn
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Lee needs a sign-up sheet for a car wash with columns for name, phone and time slot. Which instruction fits?

A. "Make it look really nice and clean."
B. "Make something people can sign up on, please."
C. "Write a story about a car wash."
D. "Make a table with columns Name, Phone and Time slot."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q1
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.PROMPT.02 (local)

Learn
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A form allows only 280 characters. Which constraint should Ana state?

A. "Use as many words as you need."
B. "Under 280 characters, including spaces."
C. "Keep it reasonably short and sweet."
D. "Make it sound exciting."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q2
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.PROMPT.02 (local)

Learn
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A coach wants practice drills a beginner can follow. Which constraint fits best?

A. "Mention every famous athlete you know."
B. "Any format is fine, whatever you like."
C. "Numbered steps, no jargon, for beginners."
D. "Make it as advanced as possible."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q3
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.PROMPT.02 (local)

Learn
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Which request would produce a result that is easiest to check?

A. "Some tips, please, as good as you can make them."
B. "Give me some tips, not too many."
C. "Write a lot of helpful tips."
D. "List exactly five tips, each under 20 words."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q4
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.PROMPT.02 (local)

Learn
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A teacher wants a parent letter without mentioning grades. How should she say this?

A. "Do not mention grades."
B. "Make it nice for parents."
C. "Use good judgment on content."
D. "Be careful about some topics."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F1-Q5
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.PROMPT.02 (local)

Learn
Format and constraints work best when they are specific and checkable. "Keep it short" is vague; "under 100 words" can be checked. "Make it look nice" is vague; "a table with columns Name, Date and Cost" can be checked. Tools do not always follow every constraint, especially exact word counts or many rules at once, so check the result against each one. If a constraint really matters, such as a character limit for a form, verify it yourself rather than trusting the tool's claim that it complied.

A response says, "Here is your summary in under 100 words," but it looks long. What should you do?

A. Trust the claim, since it said it complied
B. Count the words yourself
C. Ask it to repeat the claim
D. Assume all summaries are under 100 words

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q1
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.PROMPT.02 (local)

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A request lists twelve formatting rules, and the result breaks two. What is a sensible next step?

A. Add twelve more rules
B. Accept it without checking the rest
C. Give up on using any formatting rules from now on
D. Point out the two broken rules and ask for a fix

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q2
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.PROMPT.02 (local)

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Which version of "keep it short" can be checked?

A. "Short and snappy, please"
B. "Three sentences or fewer"
C. "Brief, as usual"
D. "Not wordy"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q3
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.PROMPT.02 (local)

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A grant form requires exactly 500 characters or fewer. The tool says its text is 480 characters. What is the safest action?

A. Check the character count before submitting
B. Ask the tool if it is sure
C. Make it longer to be safe
D. Submit it, since the tool stated the count

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q4
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.PROMPT.02 (local)

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Why might a tool miss an exact word count even when asked clearly?

A. Word counts are illegal to request
B. Tools are designed to ignore every limit on purpose
C. Exact counts are hard for it to track precisely
D. Exact counts only work in other languages

REFERENCE LABEL: NOT PART OF THE QUESTION ID S23-F2-Q5
Artificial Intelligence, Set 23: Output format and constraints · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.PROMPT.02 (local)

Set 24: Examples in a request

Outcome AI.PROMPT.03: Use a short example to show what is wanted, and notice when an example pulls the output away from the written instructions

Module: Foundation 4: Asking for what you need

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Learn
Sometimes the fastest way to explain what you want is to SHOW it. Adding one or two short EXAMPLES of the desired output, such as a sample product description or a sample quiz question, helps the tool copy the style, length and structure. But examples are powerful: the tool may copy them too closely, repeating their topic, length or wording even when your written instructions say otherwise. Use examples that match what you want, label them clearly ("Example of the style I want:"), and say what should differ ("use new topics, not the example's topic").

Why include a short example of the desired output?

A. It shows the style, length and structure to copy
B. It makes the reply arrive faster than usual
C. It proves the facts in the answer are true
D. It makes the tool ignore all other instructions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q1
Artificial Intelligence, Set 24: Examples in a request · MAIN · Q1/5 · Outcome AI.PROMPT.03 (local)

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Fictional example (prepared in advance; not a real AI output)

Request:

> Write three quiz questions about planets in the style of this example.
> Example: "Which planet is known as the Red Planet? (Mars)"

Response:

> 1. Which planet is known as the Red Planet? (Mars)
> 2. Which planet is often called the Red Planet? (Mars)
> 3. Which planet has a reddish color from iron oxide? (Mars)

In this fictional example, what went wrong?

A. The questions did not use the example's format
B. The questions were about the wrong subject entirely
C. Every question copied the example's topic, Mars
D. There were too many questions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q2
Artificial Intelligence, Set 24: Examples in a request · MAIN · Q2/5 · Outcome AI.PROMPT.03 (local)

Learn
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Fictional example (prepared in advance; not a real AI output)

Request:

> Write three quiz questions about planets in the style of this example.
> Example: "Which planet is known as the Red Planet? (Mars)"

Response:

> 1. Which planet is known as the Red Planet? (Mars)
> 2. Which planet is often called the Red Planet? (Mars)
> 3. Which planet has a reddish color from iron oxide? (Mars)

What would most likely fix the problem in that example?

A. Adding a second example that is also about Mars
B. Saying "use different planets than the example"
C. Removing the word "quiz" from the request
D. Asking for thirty questions instead of three

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q3
Artificial Intelligence, Set 24: Examples in a request · MAIN · Q3/5 · Outcome AI.PROMPT.03 (local)

Learn
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How should an example be marked in a request?

A. Written in capital letters only
B. Hidden in the middle of the request with no label
C. Placed after the reply is finished
D. Clearly labeled, like "Example of my style:"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q4
Artificial Intelligence, Set 24: Examples in a request · MAIN · Q4/5 · Outcome AI.PROMPT.03 (local)

Learn
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What is a risk of including examples?

A. The tool will refuse to answer
B. The answer will be in another language
C. The tool may copy them too closely
D. The tool will always ignore them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-M-Q5
Artificial Intelligence, Set 24: Examples in a request · MAIN · Q5/5 · Outcome AI.PROMPT.03 (local)

Learn
If you bring a TAILOR your favorite shirt and say "make one like this," the tailor copies its fit, collar and length. That is helpful. But if you wanted a DIFFERENT color and forgot to say so, you may get the same color too, because the sample shows everything, not only the parts you meant. A tailor also tends to trust the shirt in hand over a quick comment. Examples in a request work the same way: they show a lot, and the tool may copy more than you intended.

What does the tailor copy from the sample shirt?

A. Its fit, collar and length
B. Only the store it came from
C. Only its price tag
D. Nothing at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q1
Artificial Intelligence, Set 24: Examples in a request · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.PROMPT.03 (local)

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Why might you get the same color?

A. The tailor has only one color of cloth
B. You did not say the color should differ
C. The shirt was the wrong size
D. Colors cannot be changed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q2
Artificial Intelligence, Set 24: Examples in a request · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.PROMPT.03 (local)

Learn
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What tends to win: the shirt in hand or a quick comment?

A. Neither
B. The tailor's mood
C. The quick comment
D. The shirt in hand

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q3
Artificial Intelligence, Set 24: Examples in a request · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.PROMPT.03 (local)

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In a request, what is like the sample shirt?

A. The user's name
B. The tool's privacy settings
C. An example of the output
D. The tool's price

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q4
Artificial Intelligence, Set 24: Examples in a request · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.PROMPT.03 (local)

Learn
If you bring a TAILOR your favorite shirt and say "make one like this," the tailor copies its fit, collar and length. That is helpful. But if you wanted a DIFFERENT color and forgot to say so, you may get the same color too, because the sample shows everything, not only the parts you meant. A tailor also tends to trust the shirt in hand over a quick comment. Examples in a request work the same way: they show a lot, and the tool may copy more than you intended.

How do you get a different color from the tailor?

A. Ask for the same shirt twice
B. Say clearly what should differ
C. Say nothing
D. Bring the same shirt again

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-B1-Q5
Artificial Intelligence, Set 24: Examples in a request · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.PROMPT.03 (local)

Learn
Sometimes the fastest way to explain what you want is to SHOW it. Adding one or two short EXAMPLES of the desired output, such as a sample product description or a sample quiz question, helps the tool copy the style, length and structure. But examples are powerful: the tool may copy them too closely, repeating their topic, length or wording even when your written instructions say otherwise. Use examples that match what you want, label them clearly ("Example of the style I want:"), and say what should differ ("use new topics, not the example's topic").

Dana wants product descriptions like her best one, but for new products. What should she write?

A. Her best description pasted with no label or note
B. A list of products with no example at all
C. Only "write some product descriptions"
D. Her best one, labeled, plus "use the new details"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q1
Artificial Intelligence, Set 24: Examples in a request · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.PROMPT.03 (local)

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A teacher gives one sample math word problem about apples. The results all mention apples. What should she change?

A. Add a varied second example or ask for other topics
B. Nothing; apples are required in math problems
C. Give five more apple problems as examples
D. Remove all instructions and keep the example

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q2
Artificial Intelligence, Set 24: Examples in a request · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.PROMPT.03 (local)

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Which request best uses an example?

A. "Copy this exactly: 'Bright, simple.'"
B. "Write three descriptions for new products, nicely."
C. "Style example: 'Bright, simple.' Write three for new products."
D. "Write three product descriptions in a style I would like."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q3
Artificial Intelligence, Set 24: Examples in a request · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.PROMPT.03 (local)

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Why label an example clearly?

A. So it is not mistaken for an instruction
B. So the reply becomes longer
C. So the tool can skip reading the example
D. So the facts become verified

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q4
Artificial Intelligence, Set 24: Examples in a request · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.PROMPT.03 (local)

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An example email is formal, but you want a casual email. What is the simplest fix?

A. Keep the formal example and say nothing else
B. Ask for "an email" with no example
C. Add three more formal examples
D. Use a casual example, or describe the tone

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F1-Q5
Artificial Intelligence, Set 24: Examples in a request · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.PROMPT.03 (local)

Learn
When written instructions and examples disagree, the example often wins. If your instructions say "write four-sentence descriptions" but your only example has two sentences, expect short results. If every example mentions the same product, the tool may keep mentioning it. Fix conflicts by changing the example to match the rules, giving two varied examples instead of one, or removing the example and describing the pattern in words. Then check that results follow the rules, not just the example.

Instructions say "four sentences each," but the only example has two sentences. What result is likely?

A. Descriptions in a different language
B. Short descriptions that follow the example
C. No descriptions at all
D. Exactly four sentences every time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q1
Artificial Intelligence, Set 24: Examples in a request · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.PROMPT.03 (local)

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Every example mentions "Brand X." The new descriptions keep mentioning Brand X. What is the best fix?

A. Use examples without Brand X or vary them
B. Add more examples with Brand X
C. Remove the instructions but keep the examples
D. Ask for longer descriptions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S24-F2-Q2
Artificial Intelligence, Set 24: Examples in a request · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.PROMPT.03 (local)

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After fixing a conflicting example, what should you still check?

A. Nothing; fixing the example guarantees success
B. Only that results match the example's topic
C. That results follow the written rules
D. Only the number of words in the request

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When might it be better to remove an example and describe the pattern in words?

A. Never; examples are always required
B. When the example matches the rules perfectly
C. When you want results to copy the example
D. When the example keeps pulling results the wrong way

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Two varied examples instead of one usually help because they...

A. guarantee every fact is correct
B. show what is the same and what can vary
C. force the tool to ignore instructions
D. make the request use fewer tokens

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